@meteatamel

Containers, Kubernetes and Google Cloud

Mete Atamel

Developer Advocate for Google Cloud

Google Cloud Platform

Mete Atamel

Developer Advocate for Google Cloud

@meteatamel atamel@google.com

meteatamel.wordpress.com

Please send talk feedback: bit.ly/atamel

Software development is HARD!

And it is not getting any easier…

In the good old days

This is all I had to care...

Life was good :-)

Image Source: Wikipedia

A lot happened since then

Internet

Databases

The Monolith

Mobile

App Servers

Web Servers

Microservices

Cloud Computing

Firewalls

Version Control

Caching

Machine Learning

Object Oriented Programming

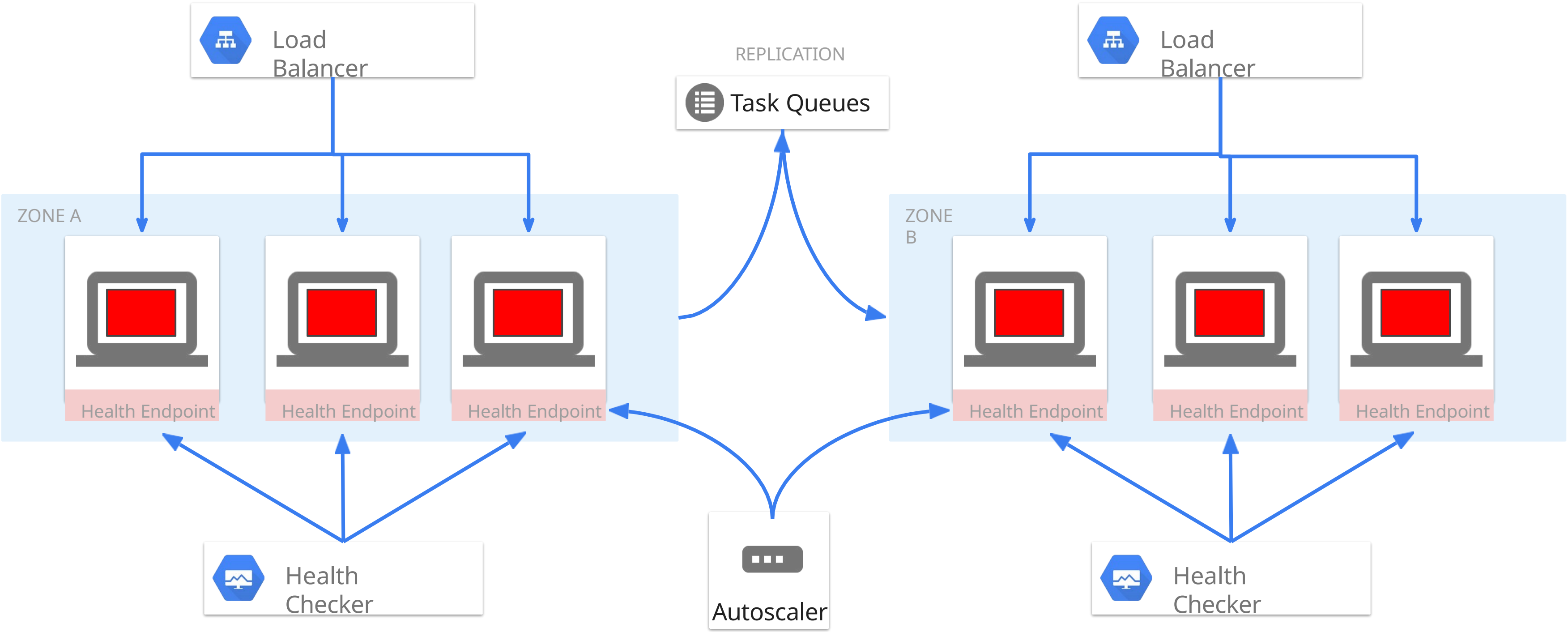
DevOps

Virtual Machines

Big Data

IoT

Nowadays


Load Balancer
REPLICATION
Task Queues
Load Balancer
ZONE A
Health Endpoint
Health Endpoint
Health Endpoint
ZONE B
Health Endpoint
Health Endpoint
Health Endpoint
Health Checker
Autoscaler
Health Checker

We haven’t even talked about

Maintaining code in **different** languages on **different** types of machines

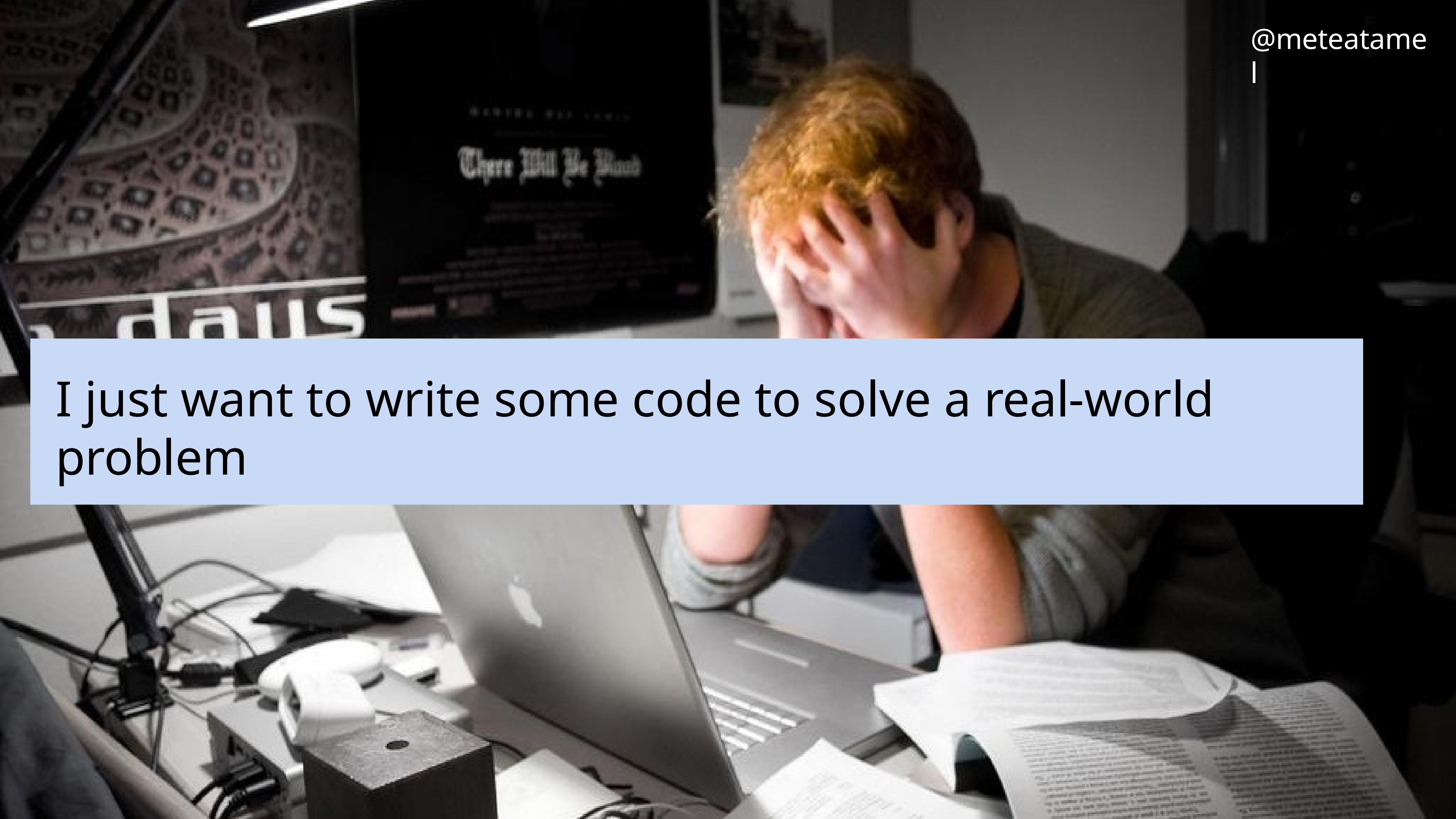
Rolling out the **new version** of your code reliably

Rolling back to the **old version** if something goes

wrong Managing **configuration** and **secrets**

Managing **scripts** that need to run on each machine

@meteatamel



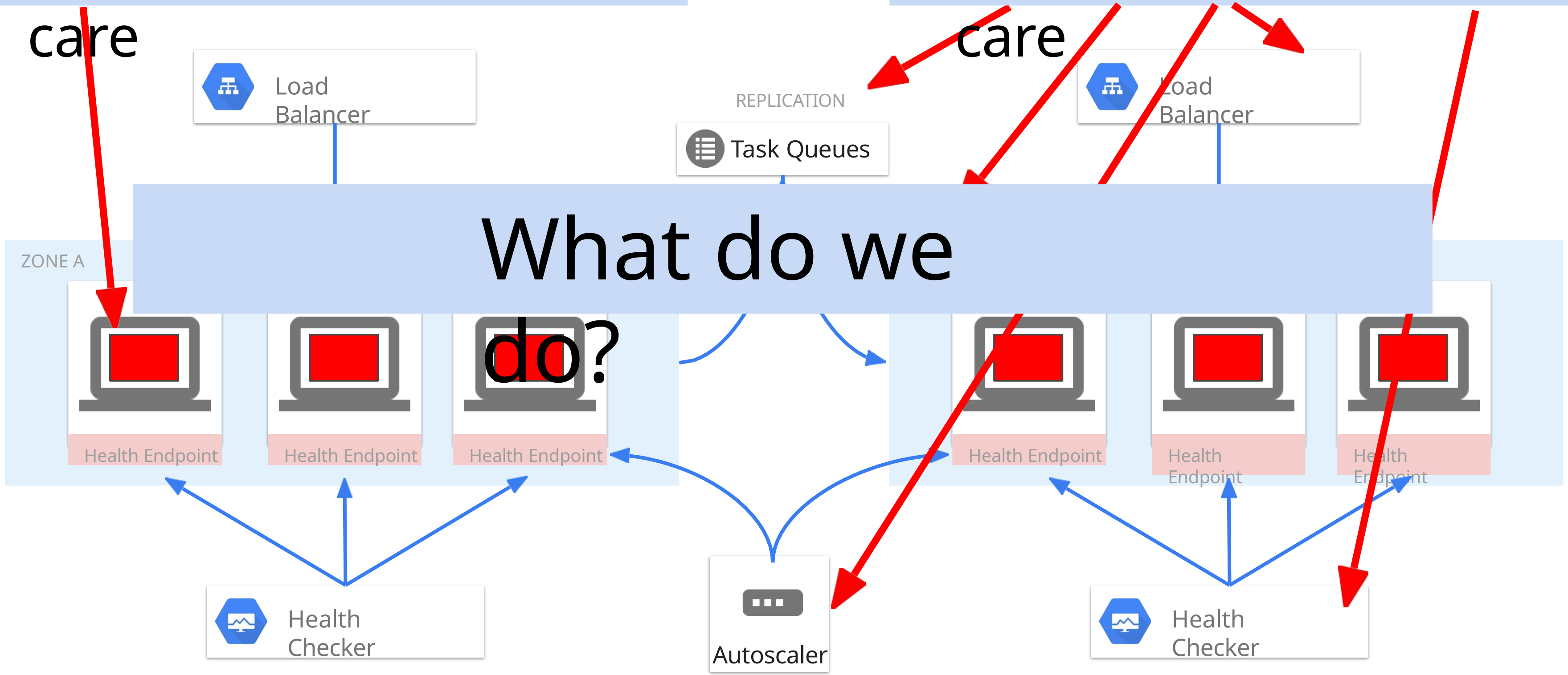
I just want to write some code to solve a real-world problem

The reality

This is all I **want** to care

This is all I **have** to care

What do we do?



In the good old days

Write your code, pass it to QA for testing, let operations team run it...

Nowadays, it is your problem

What do we do?

@meteatame
l

You can write your code in **any** language and run
anywhere exactly the **same** way

Your app is optimally deployed **somewhere** and managed by **someone.** It just works!

There are no machines. All resources are **automatically** provisioned on demand

Docker + Kubernetes + Cloud

Write your code in any language and run it anywhere exactly the same way

~ Containers (eg. Docker, Rkt)

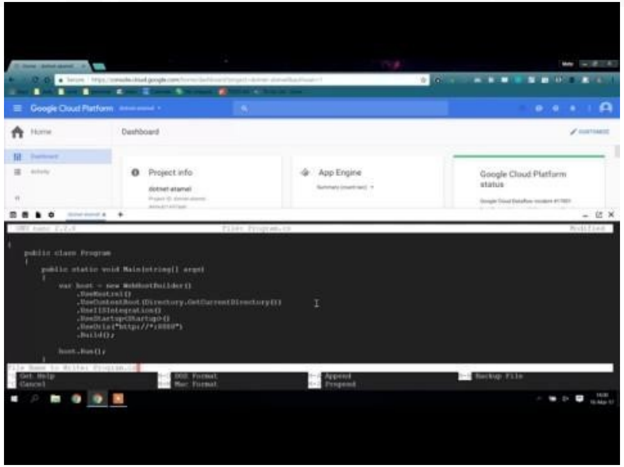
Your app is optimally deployed and managed

~ Container Management Platforms (eg. Kubernetes, Docker Swarm, Mesos)

All the resources needed for your app is automatically provisioned per demand

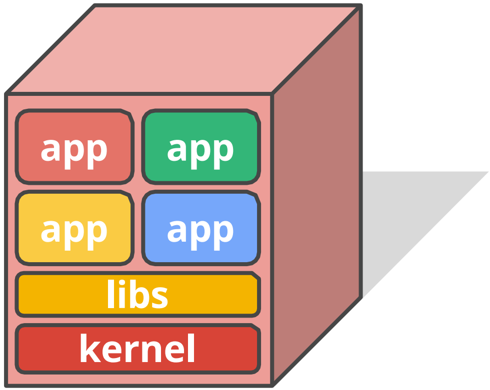
~ Cloud Providers (eg. Google Cloud, AWS, Azure)

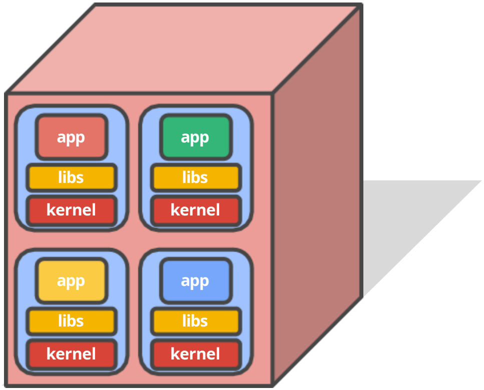
Demo: Simple

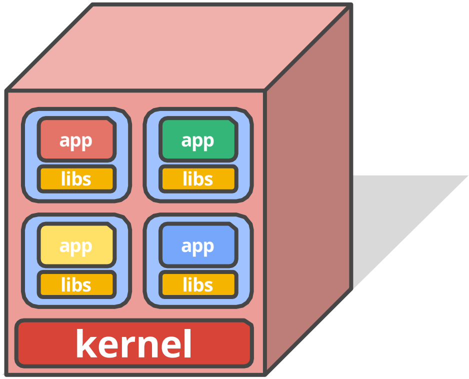

Google Cloud Platform
Home
Dashboard
Project info
App Engine
Google Cloud Platform status

Container

Why containers?







Physical Machine

- ✗ No isolation
- ✗ Common libs
- ✗ Highly coupled Apps & OS

Virtual Machines

- ✓ Isolation
- ✓ No Common Libs
- ✗ Expensive and Inefficient
- ✗ Hard to manage

Containers

- ✓ Isolation
- ✓ No Common Libs
- ✓ Less overhead
- ✗ Less Dependency on Host OS

What is a container?

Docker

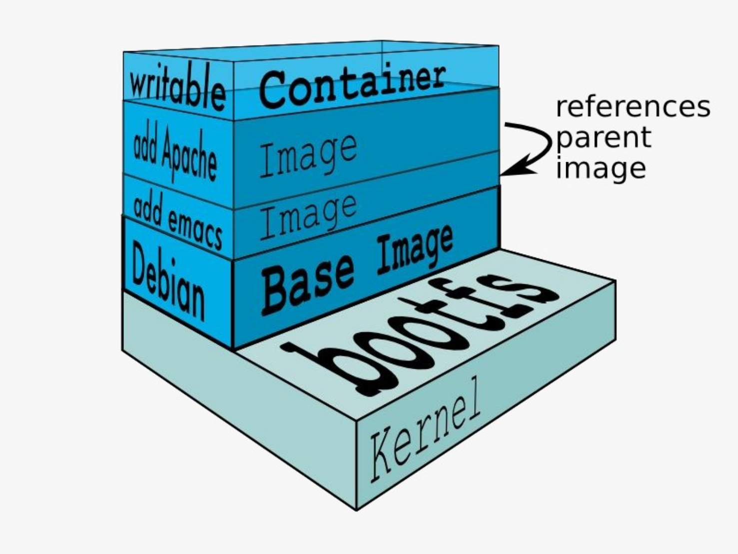
rkt

A **lightweight** way to virtualize applications

Linux (or Windows) processes

Lightweight
Hermetically sealed Isolated

Easily deployable
Introspectable
Composable



@meteatamel

Google has been developing and using containers to manage our applications for **over 12 years.**

Images by Connie Zhou

Everything at Google runs in containers

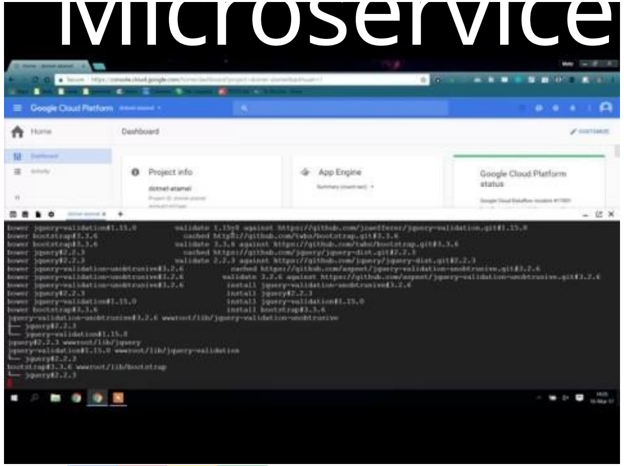
Gmail, Web Search,
Maps, ... MapReduce,
batch, ...
GFS, Colossus, ...
Even **Google's Cloud Platform:** our VMs run in containers!

We launch over **2 billion** containers **per week**

Shipping Containers At Clyde, by Steve Gibson

@meteatame
l

Demo: Containerised Microservice



Containers not enough

Containers help to create a lightweight and consistent environment for apps

But you still need to answer these questions:

- Who takes care of **redundancy**?
- Who takes care of **resiliency**?
- Who **scales up/down** your app?
- Who and how a **new version** of your app gets deployed?
- Who rolls back to a **previous version** if something goes wrong?
- Etc. etc. etc.

Kubernete

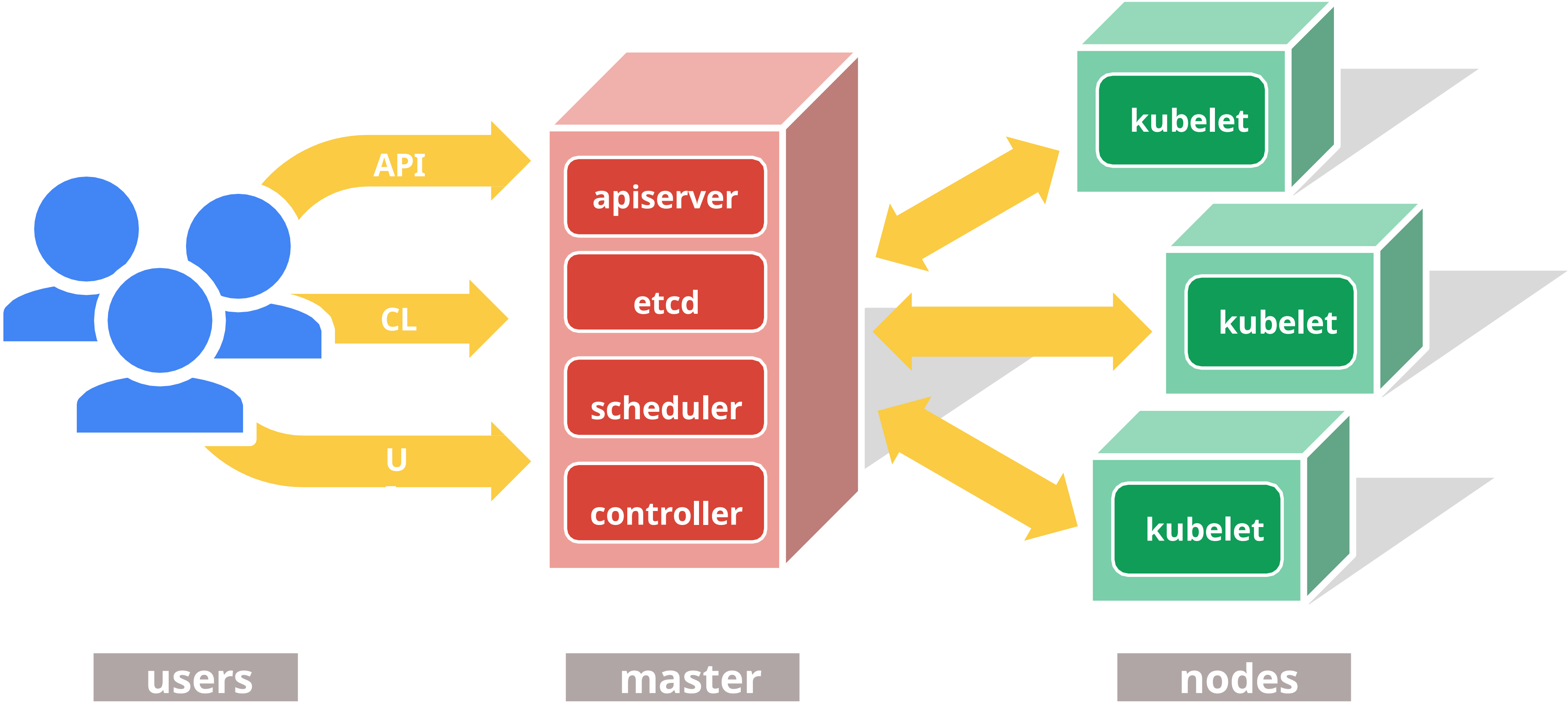
Kubernetes

Greek for *"Helmsman"*; also the root of the words *"governor"* and *"cybernetic"*

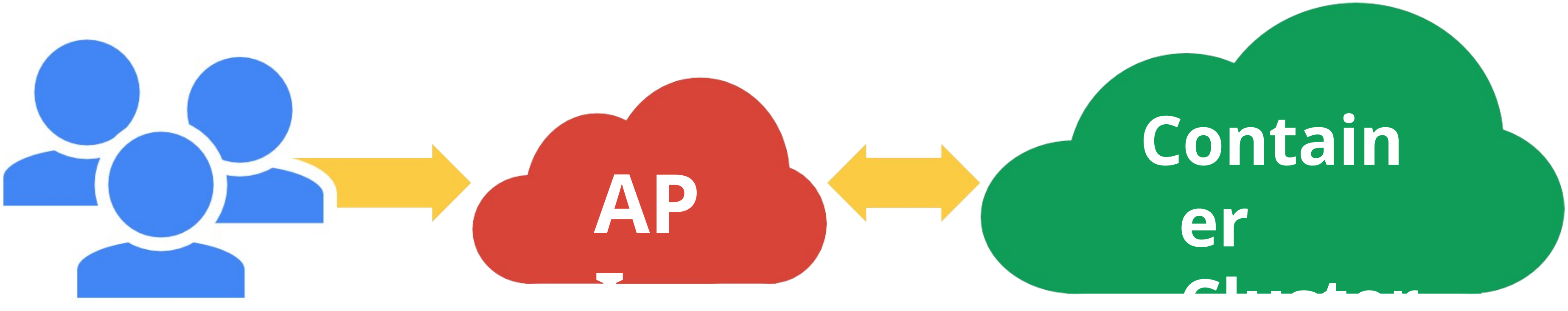
- Manages container clusters
- Inspired and informed by Google's experiences and internal systems (borg)
- Supports multiple cloud and bare-metal environments
- Supports multiple container runtimes
- **100% Open source**, written in Go

The 10000 foot view

@meteatame
l



All you really care



Container clusters: A story in two parts

@meteatamel

1. Setting up the cluster

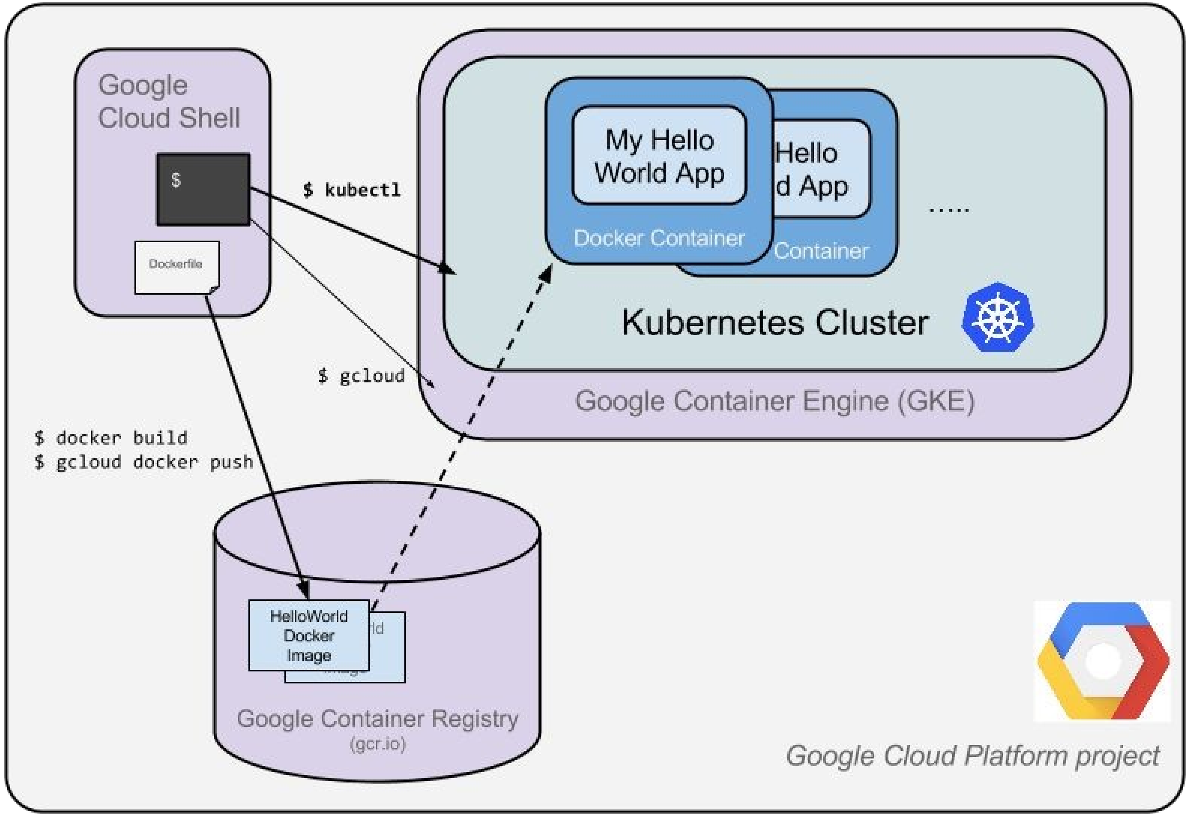
- Choose a cloud: GCE, AWS, Azure, Rackspace, on-premises, ...
- Choose a node OS: CoreOS, Atomic, RHEL, Debian, CentOS, Ubuntu, ...
- Provision machines: Boot VMs, install and run kube components, ...
- Configure networking: IP ranges for Pods, Services, SDN, ...
- Start cluster services: DNS, logging, monitoring, ...
- Manage nodes: kernel upgrades, OS updates, hardware failures...

Not the easy or fun part, but unavoidable

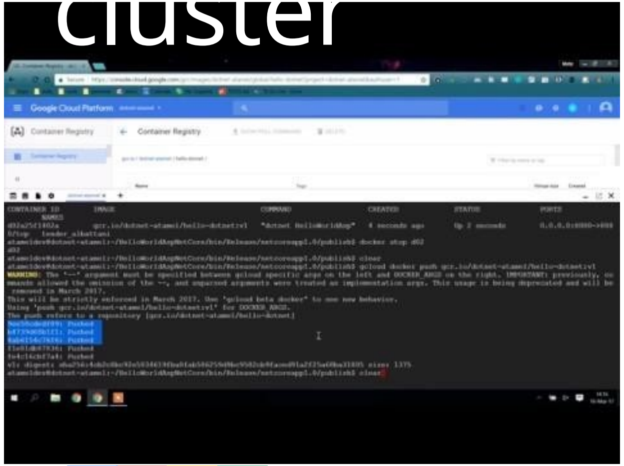
This is where things like **Google Container Engine (GKE)** really help

Kubernetes cluster on GKE

@meteatamel



Demo: Create Kubernetes cluster



Container clusters: A story in two parts

@meteatamel

2. Using the cluster

- Run Pods & Containers
- Replica Sets
- Services
- Volumes

This is the fun part!

A distinct set of problems from cluster setup and management Don’t make developers deal with cluster administration!

Accelerate development by focusing on the applications, ...

Kubernetes Building Blocks

Kubernetes Terminology

Deployment

Pod

Volume

Label

Selector

ReplicaSet

Liveness Probe

Readiness Probe

Service

DaemonSet

Job

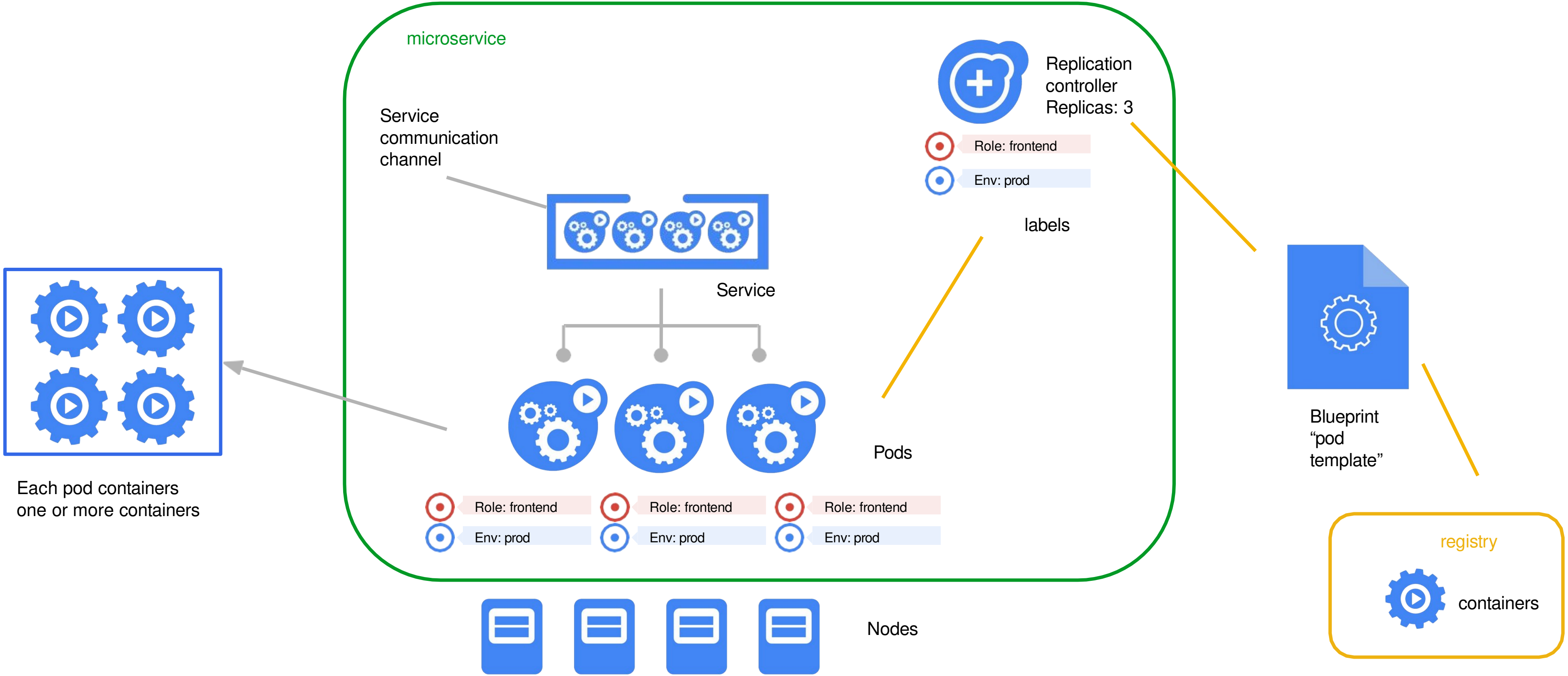
StatefulSet

ConfigMap

Secret

@meteatamel

Container cluster



Deployments: where everything

Deployments

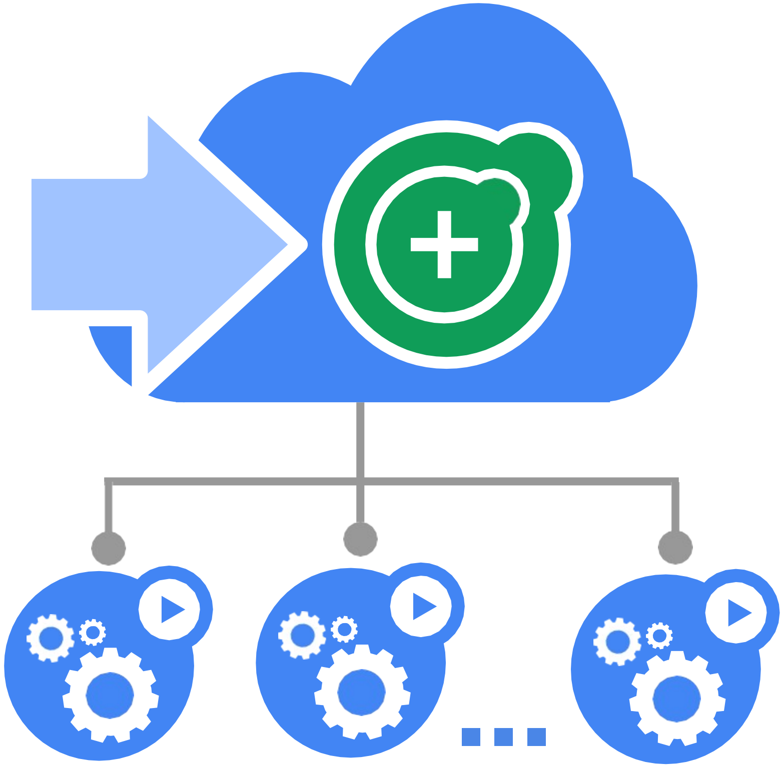
@meteatame
l

A Deployment provides declarative updates for Pods and Replica Sets

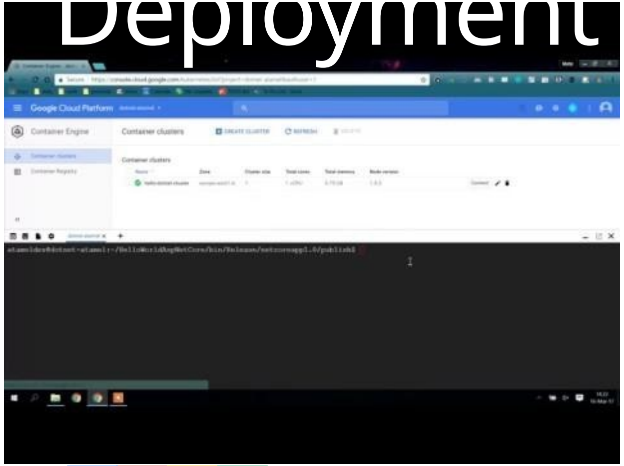
Describe the desired state and the Deployment controller will change the actual state to the desired state at a controlled rate for you.

Deployment manages replica changes for you

- stable object name
- updates are configurable, done server-side
- `kubectl edit` or `kubectl apply`



Demo: Create Deployment



Kubernetes Terminology

Deployment	ReplicaSet	DaemonSet
Pod	Liveness Probe	Job
Volume	Readiness Probe	StatefulSet
Label	Service	ConfigMap
Selector		Secret

Pods and Volumes: Atom of

Pods

Small group of containers & volumes

Tightly coupled

The atom of scheduling & placement

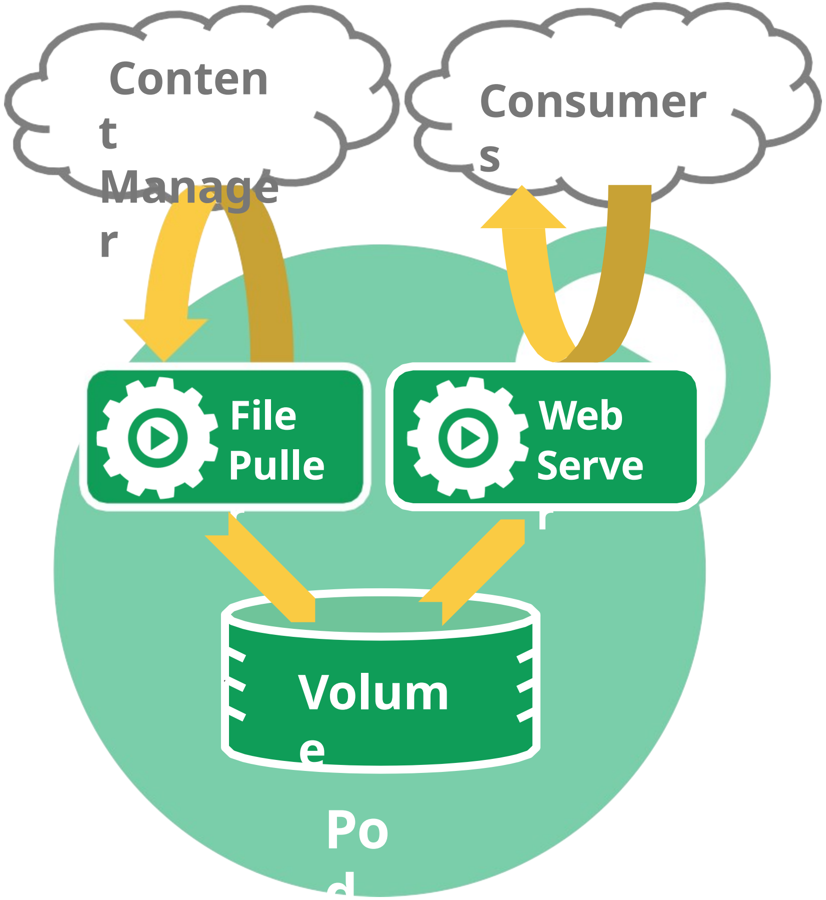
Shared namespace

- share IP address & localhost
- share IPC, etc.

Managed lifecycle

- bound to a node, restart in place
- can die, cannot be reborn with same ID

Example: data puller & web server



Volumes

Pod-scoped storage

Support many types of volume plugins

- Empty dir (and tmpfs)
- Host path
- Git repository
- GCE Persistent Disk
- AWS Elastic Block Store
- Azure File Storage
- iSCSI
- Flocker
- NFS
- vSphere
- GlusterFS
- Ceph File and RBD
- Cinder
- FibreChannel
- Secret, ConfigMap, DownwardAPI
- Flex (exec a binary)
- …

Kubernetes Terminology

Deployment	ReplicaSet	DaemonSet
Pod	Liveness Probe	Job
Volume	Readiness Probe	StatefulSet
Label	Service	ConfigMap
Selector		Secret

Labels & Selectors:

Labels

Arbitrary metadata

Attached to any API

object Generally

represent **identity**

Queryable by **selectors**

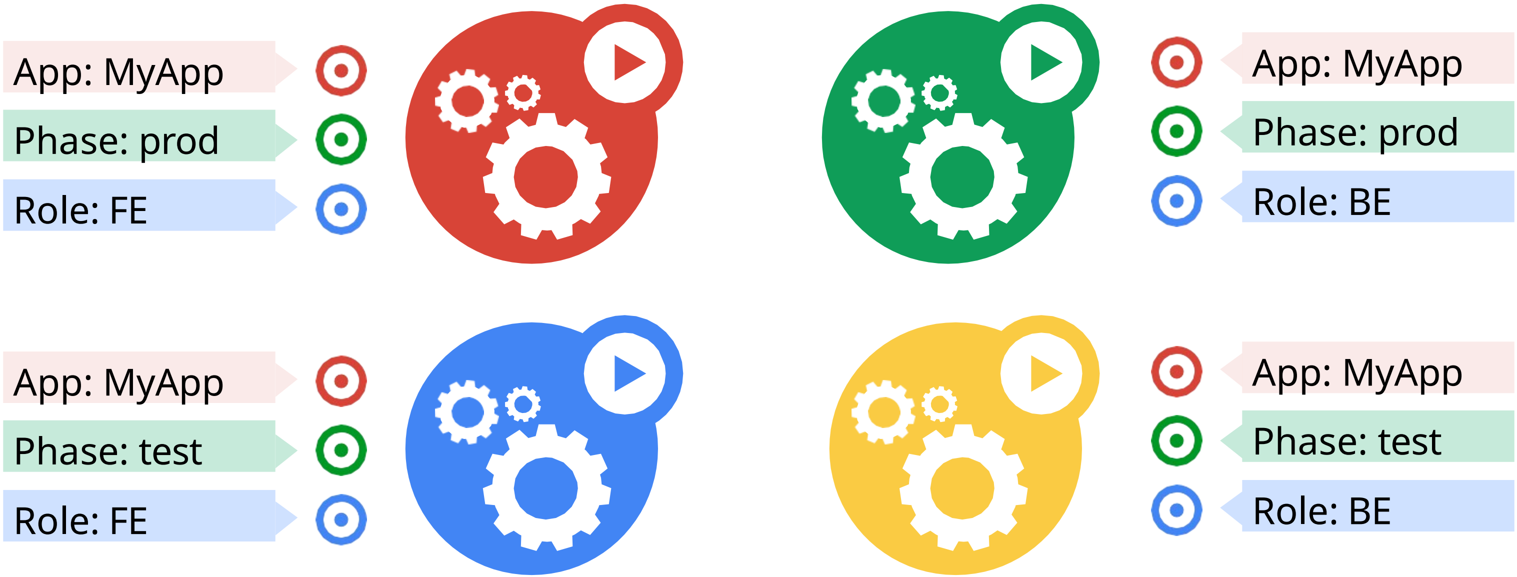
- think SQL *'select ... where ...'*

The **only** grouping mechanism

- pods under a ReplicationController
- pods in a Service
- capabilities of a node

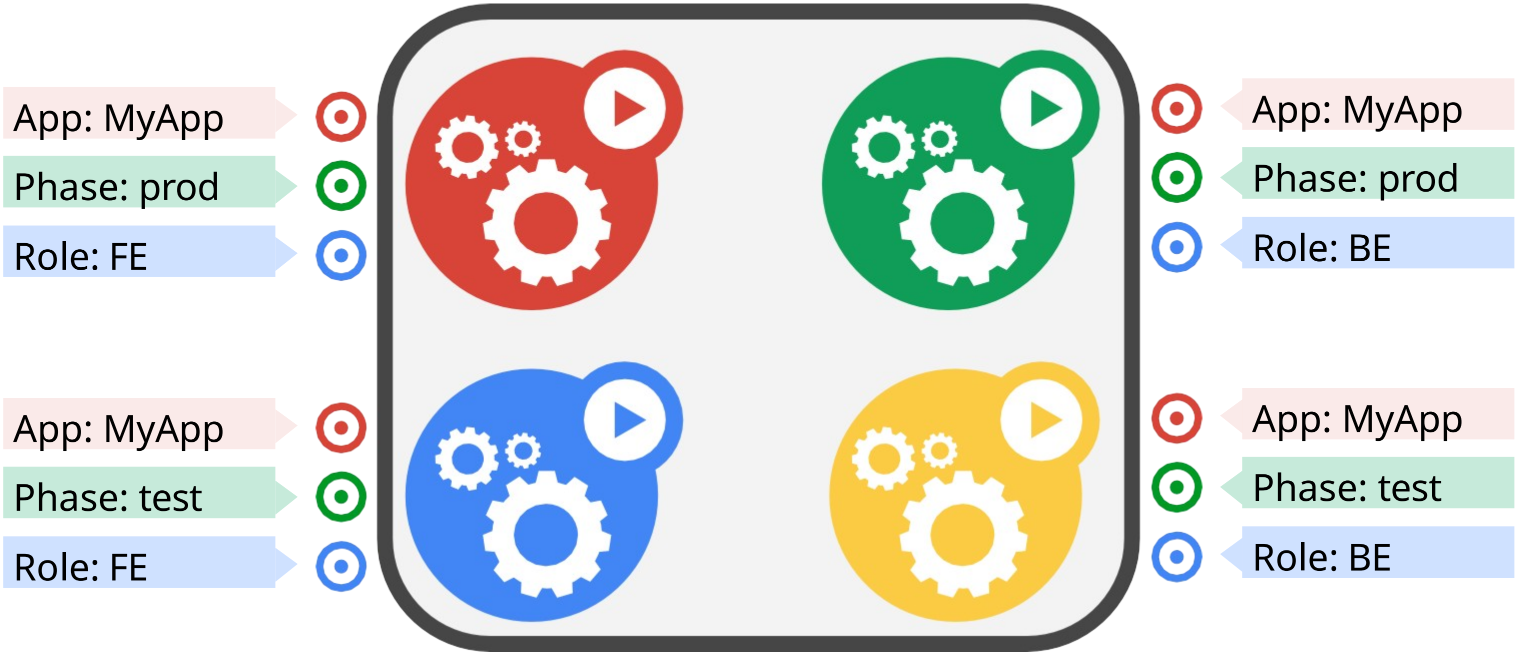
Selectors

@meteatame
l



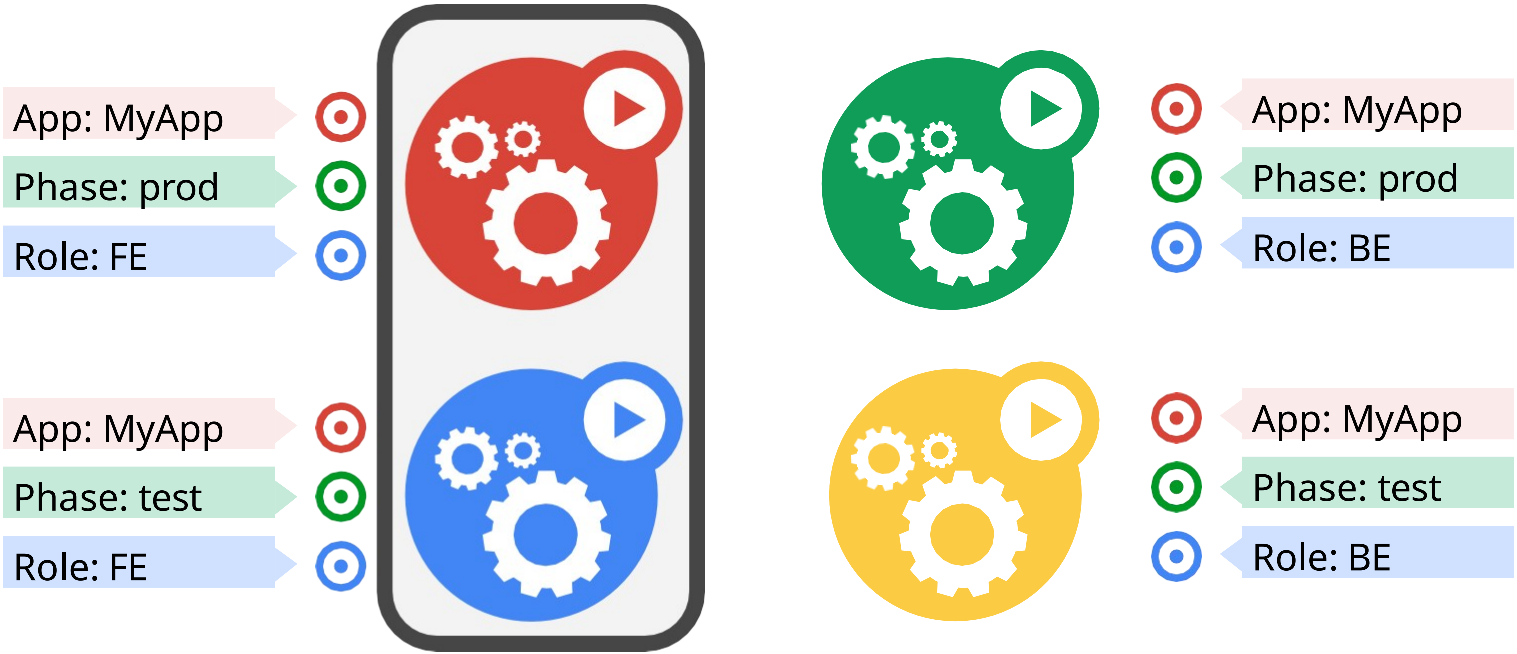
Selectors

@meteatame
l



App = MyApp

Selectors

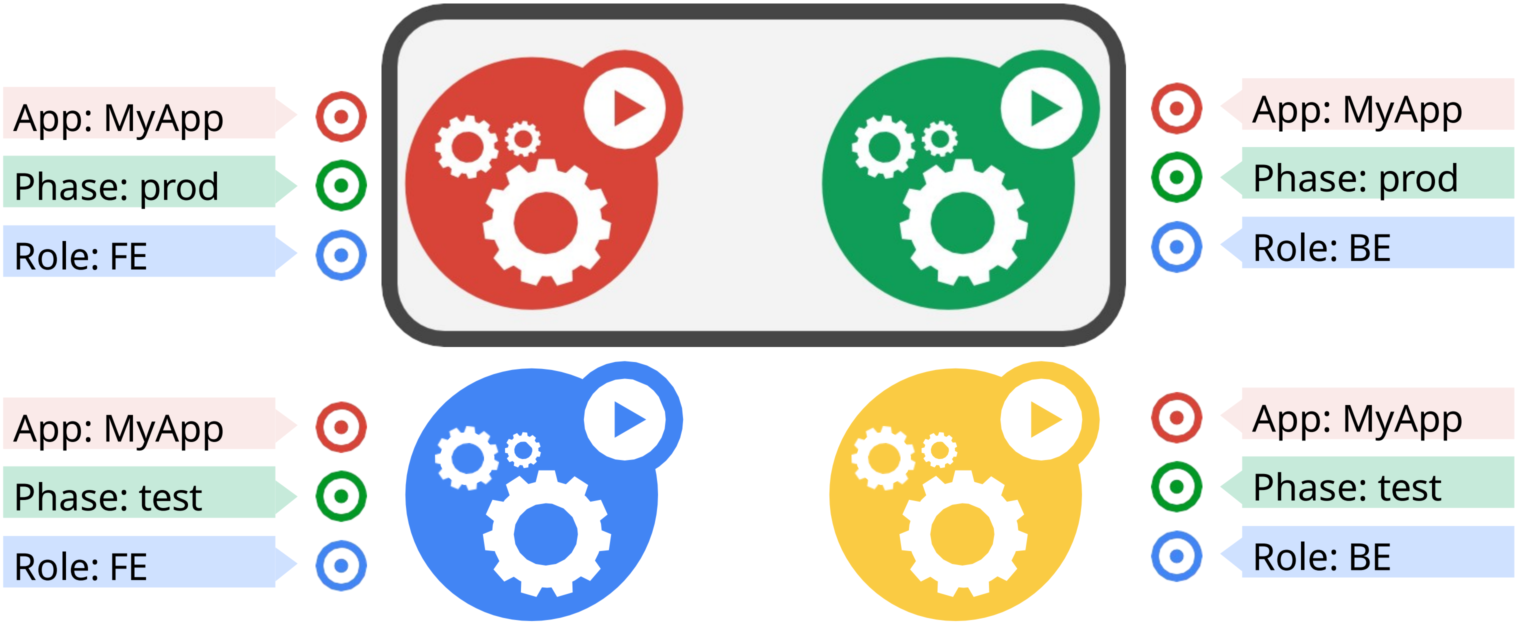


App = MyApp, Role = FE

Selectors

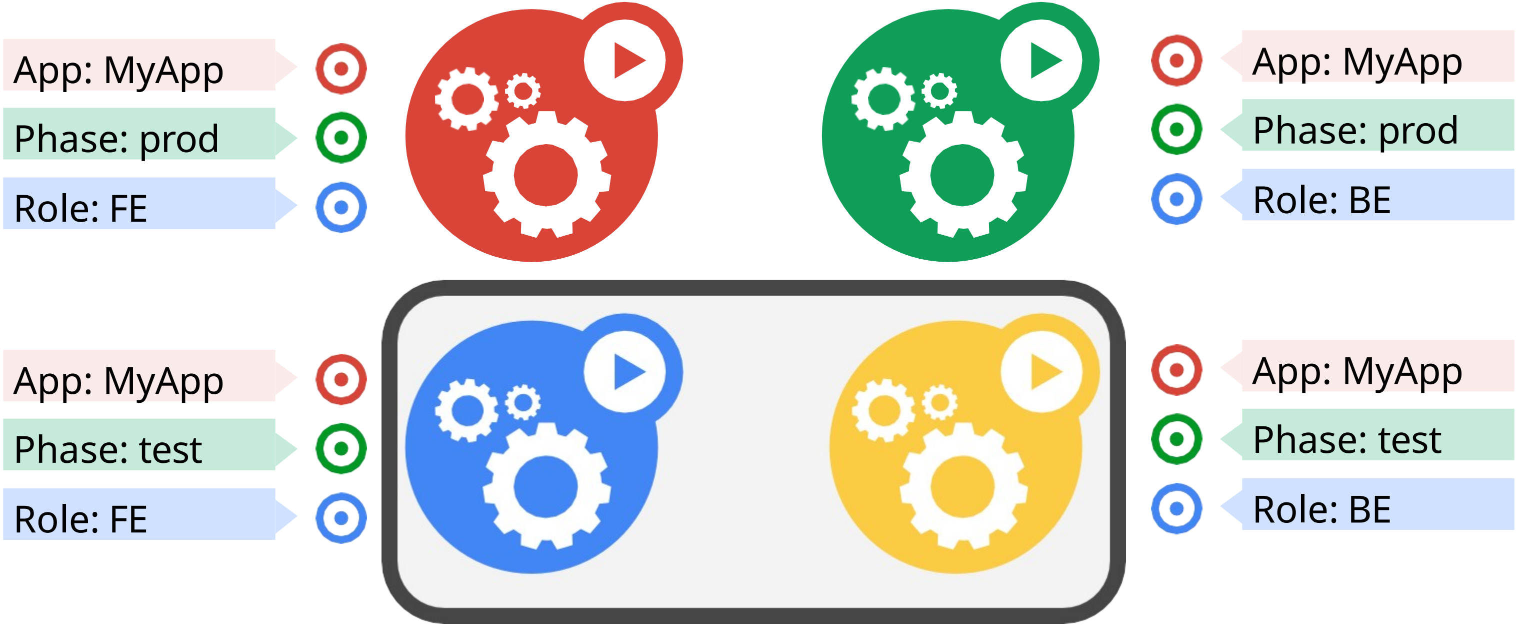
App = MyApp, Role = BE

Selectors



App = MyApp, Phase = prod

Selectors



App = MyApp, Phase = test

Kubernetes Terminology

Deployment	ReplicaSet	DaemonSet
Pod	Liveness Probe	Job
Volume	Readiness Probe	StatefulSet
Label	Service	ConfigMap
Selector		Secret

Resiliency &

ReplicaSets: Big brother

A simple control loop

Runs out-of-process wrt API server

One job: ensure N copies of a pod

grouped by a selector

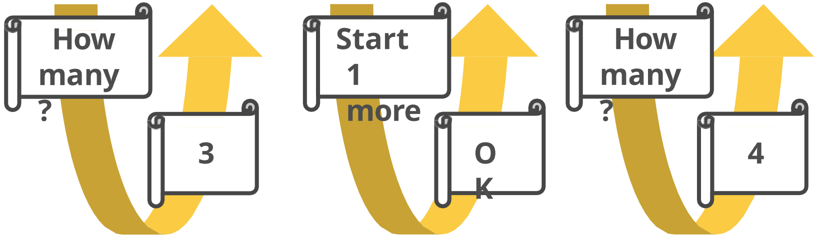
- too few? start some
- too many? kill some

Layered on top of the public Pod API

Replicated pods are **fungible**

- No implied order or identity

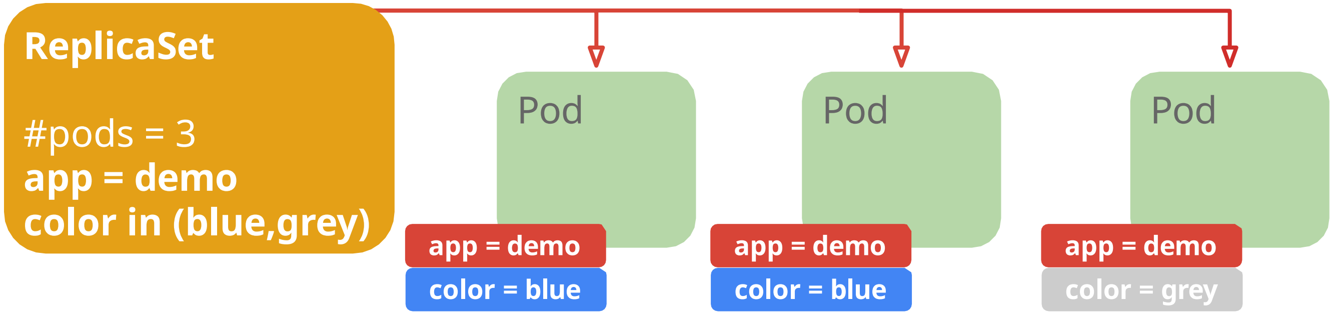
ReplicaSet
- **name = “my-rc”**
- **selector = {“App”: “MyApp”}**
- **template = { ... }**



API Server

* The evolution of ReplicationControllers

ReplicaSets



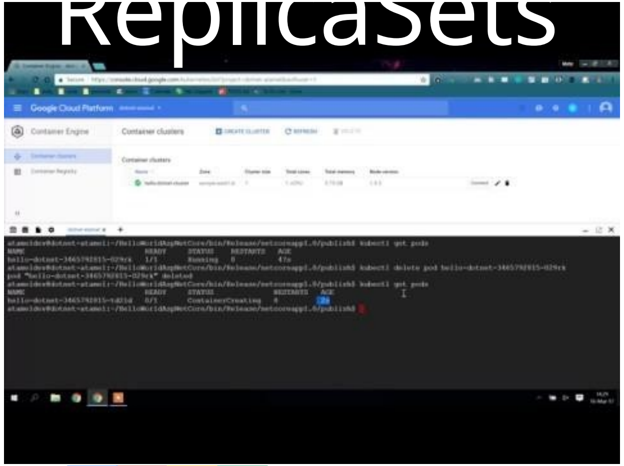
Behavior

- Keeps Pods running
- Gives direct control of Pod #s
- Grouped by Label Selector

Benefits

➔ Recreates Pods, maintains desired state
➔ Fine-grained control for scaling
➔ Standard grouping semantics

Demo: ReplicaSets



Kubernetes Health Checks

Health Check Philosophy

It's **your** responsibility to let Kubernetes know whether your app is healthy or not!

Liveness Probes

Liveness Probes make sure your application is running

livenessProbe:
 # an http probe
 httpGet:
  path: /healthz
  port: 8080
 initialDelaySeconds: 15
 timeoutSeconds: 1

wait 15 seconds after pod is started to check for health # wait 1 second for a response to health check

Readiness Probes

Readiness probes make sure your application is ready to serve traffic

readinessProbe:
  # an http
  probe
  httpGet:
    path:
    /readiness
    port: 8080
  initialDelaySeconds: 20 # wait 20 seconds after pod is started to check for
  health timeoutSeconds: 5   # wait 5 second for a response to health check

Kubernetes Terminology

Deployment	**ReplicaSet**	DaemonSet
Pod	**Liveness Probe**	Job
Volume	**Readiness Probe**	StatefulSet
Label	Service	ConfigMap
Selector		Secret

Service

Services

A logical grouping of pods that perform the same function (the Service's endpoints)

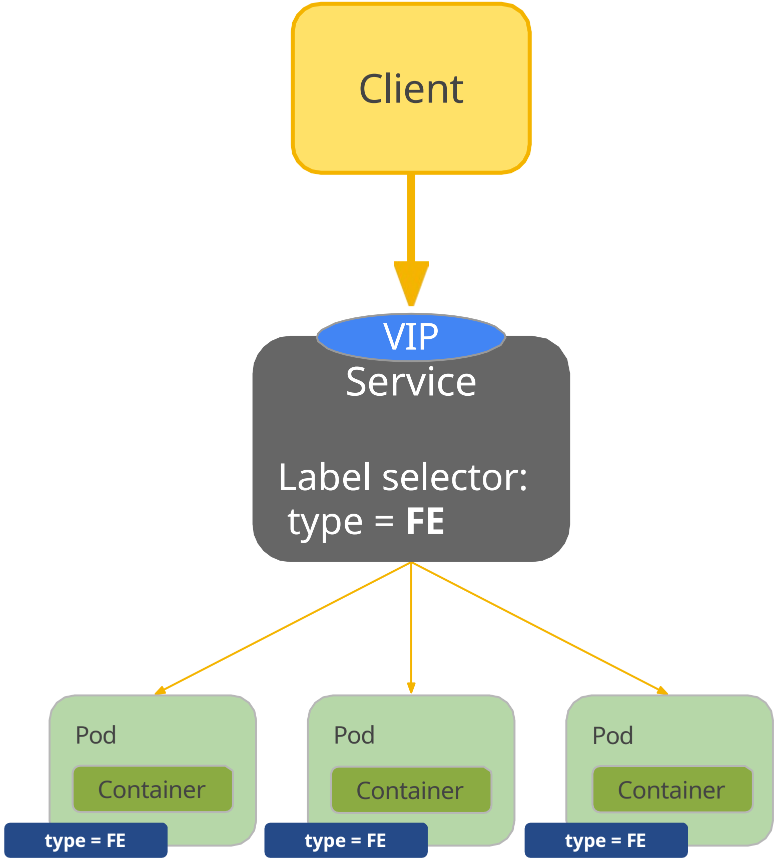
- grouped by label selector

Load balances incoming requests across constituent pods

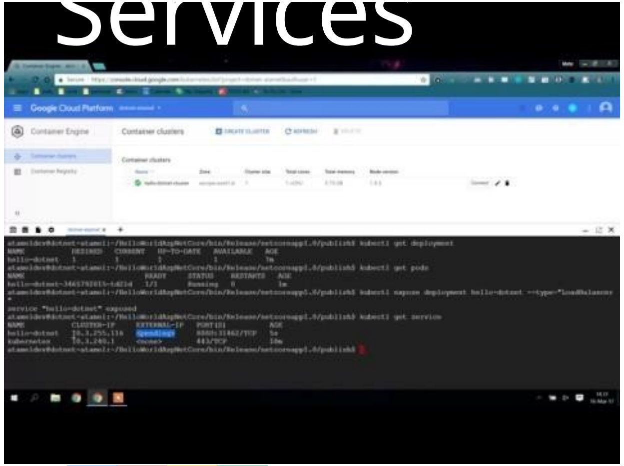
Choice of pod is random but supports session affinity (ClientIP)

Gets a **stable** virtual IP and port

- also a DNS name

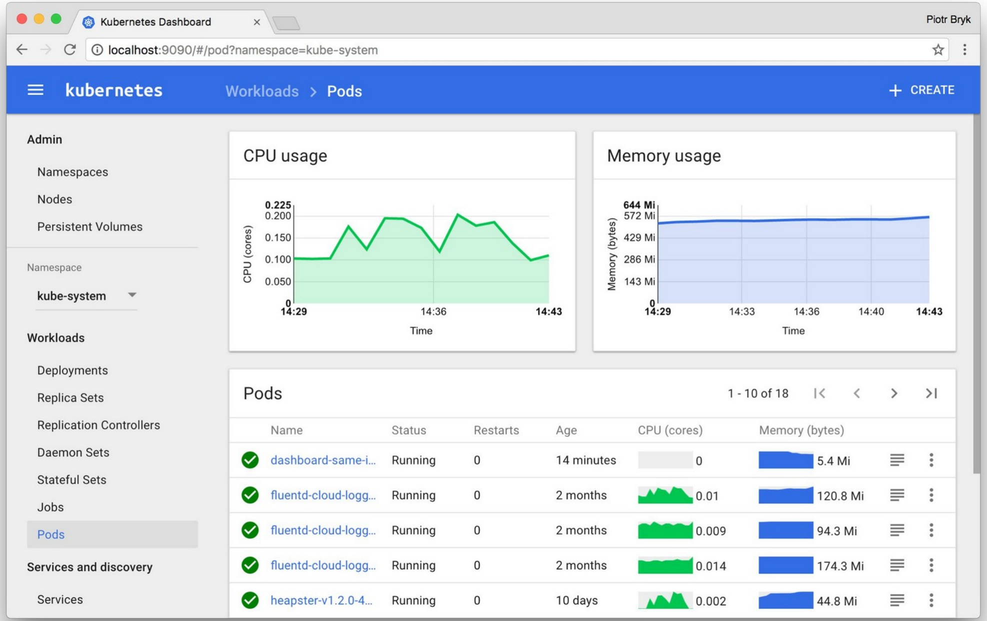


Demo: Services

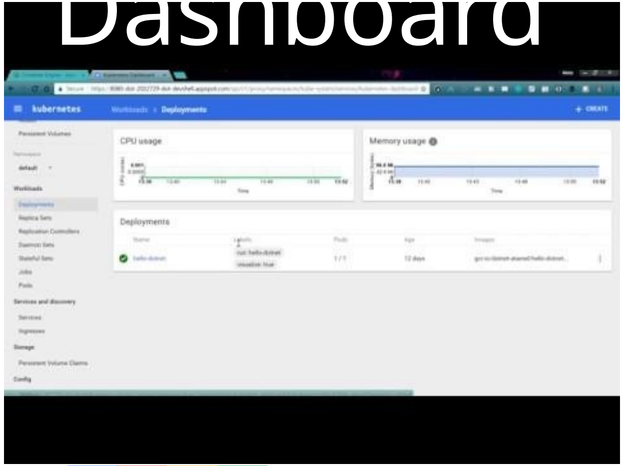


Kubernetes Dashboard

A general purpose, web-based UI to view/manage Kubernetes clusters



Demo: Kubernetes Dashboard

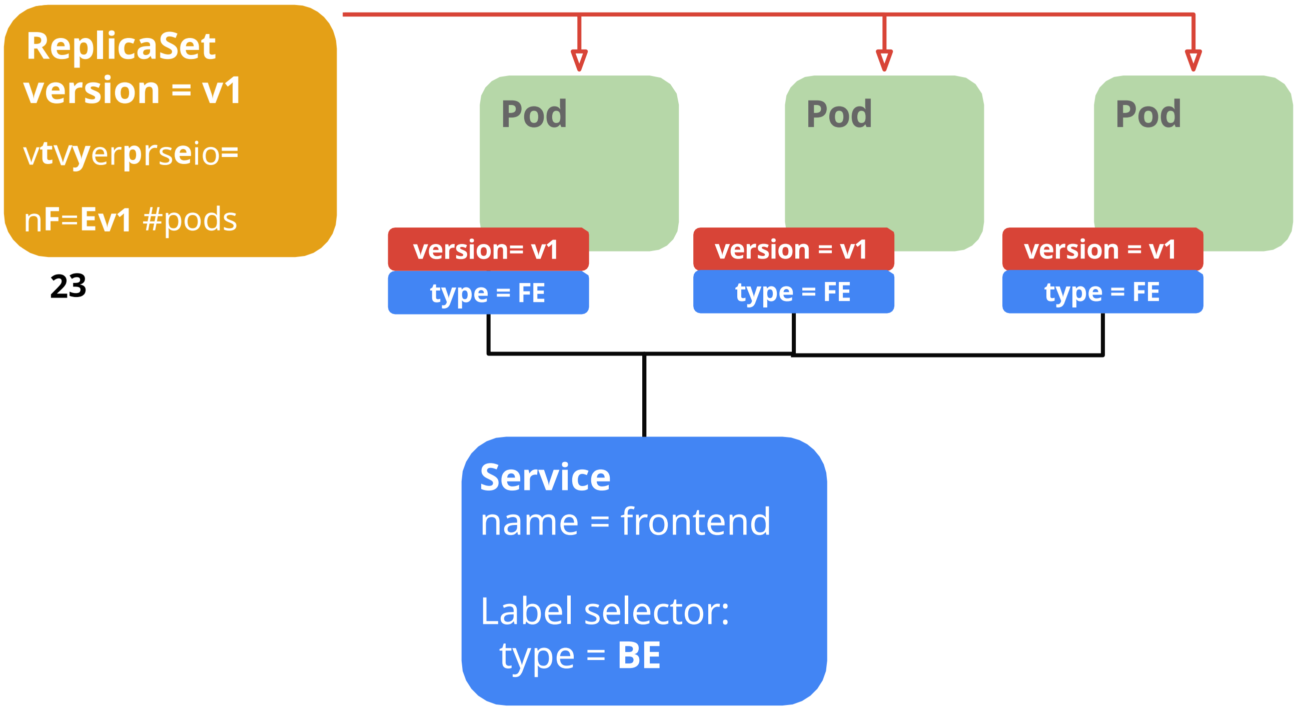


Kubernetes Terminology

Deployment	**ReplicaSet**	DaemonSet
Pod	**Liveness Probe**	Job
Volume	**Readiness Probe**	StatefulSet
Label	**Service**	ConfigMap
Selector		Secret

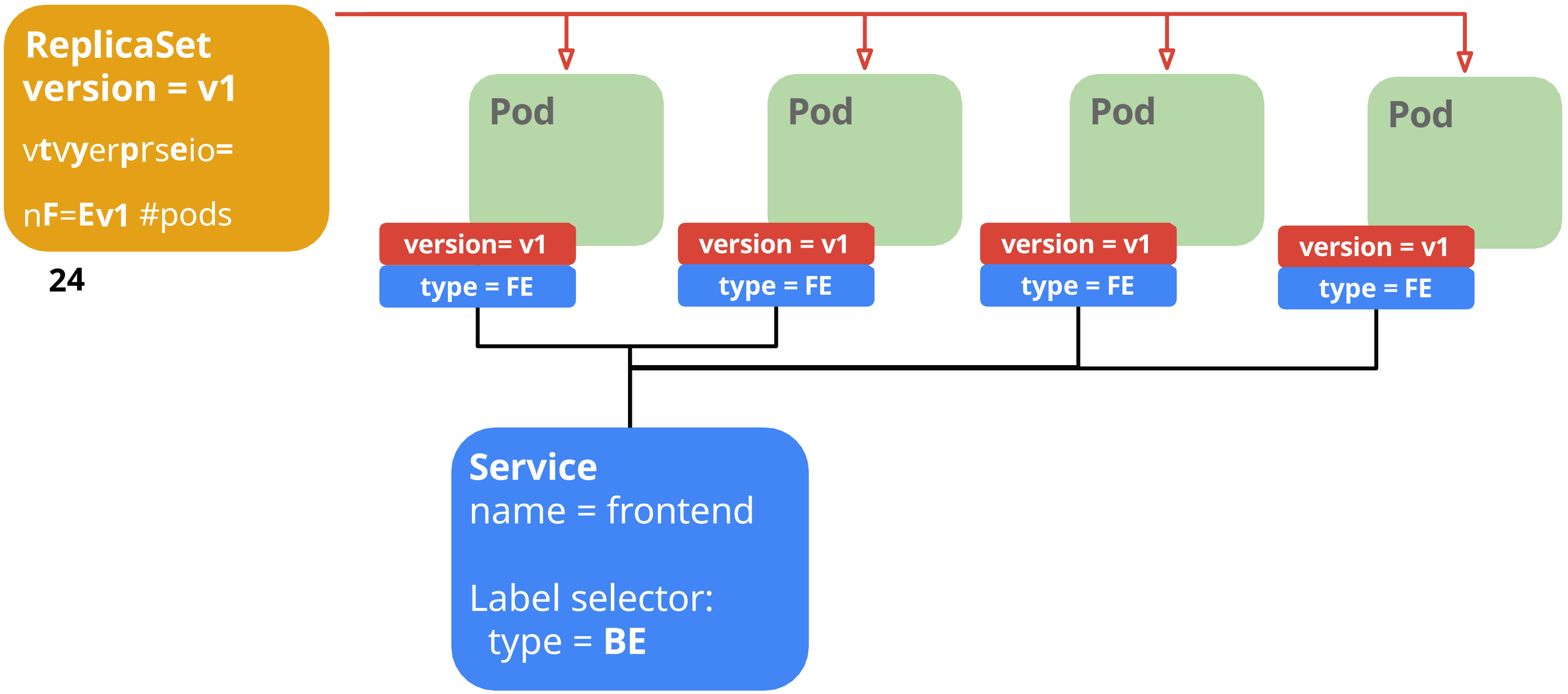
Scalin

Scaling

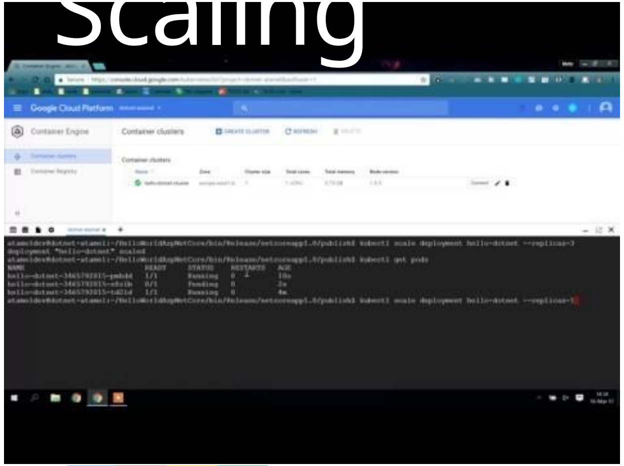

ReplicaSet
version = v1
vtvyerprseio=
nF=Ev1 #pods
23
Pod
Pod
Pod
version= v1
type = FE
version = v1
type = FE
version = v1
type = FE
Service
name = frontend
Label selector:
type = BE

Scaling

@meteatame
l



Demo: Scaling

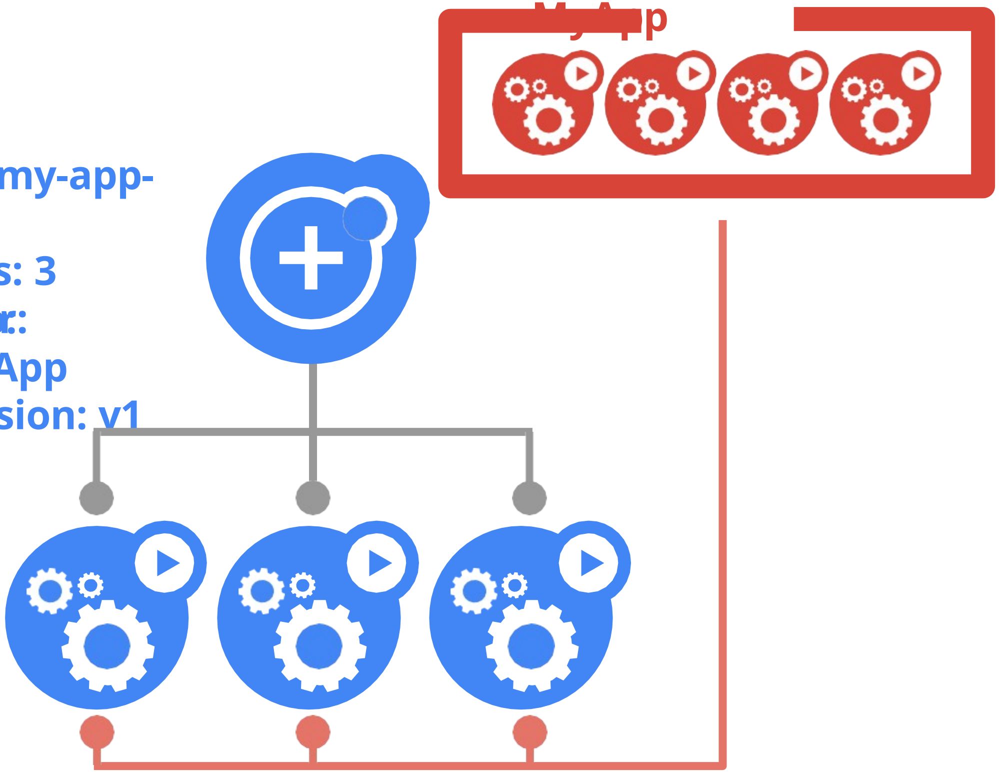


Rolling

Rolling Update

Service
- **app: MyApp**

ReplicaSet
- **name: my-app-v1**
- **replicas: 3**
- **selector:**
  - **app: MyApp**
  - **version: v1**

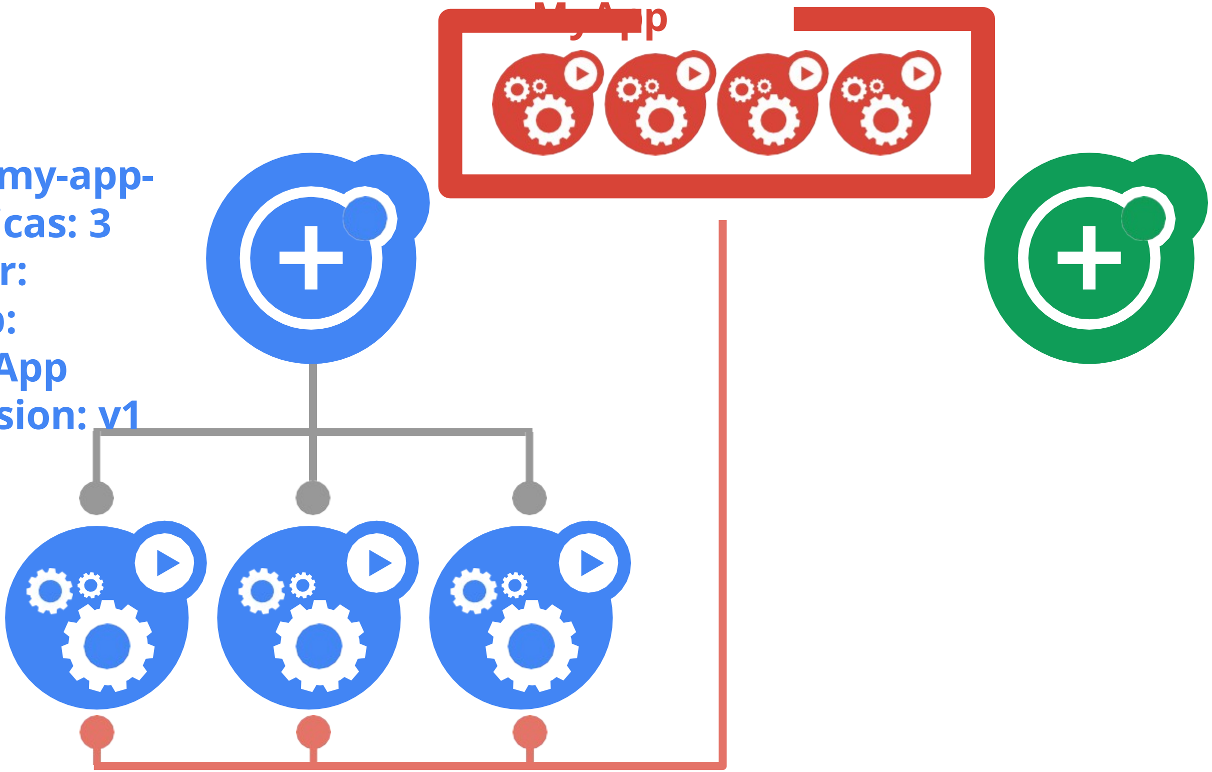


Rolling Update

@meteatamel

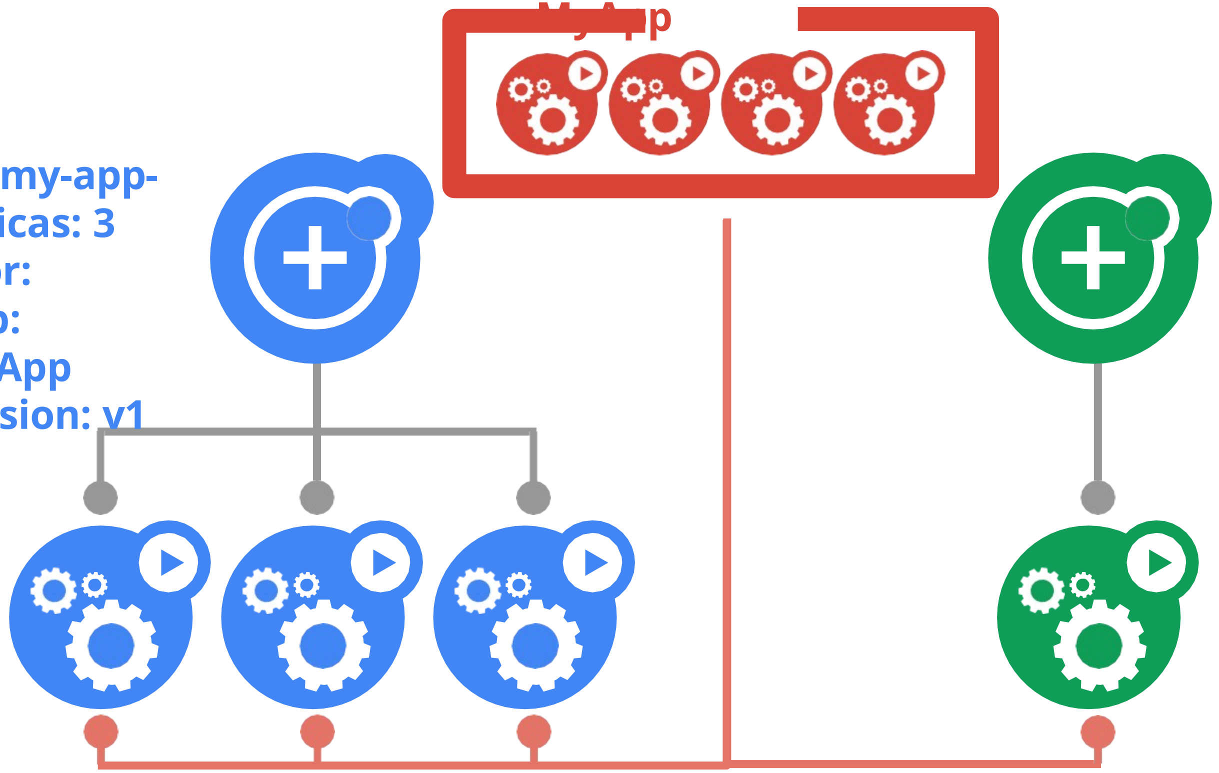
Service
- app: MyApp

ReplicaSet
- name: my-app-v1
- replicas: 3
- selector:
  - app: MyApp
  - version: v1



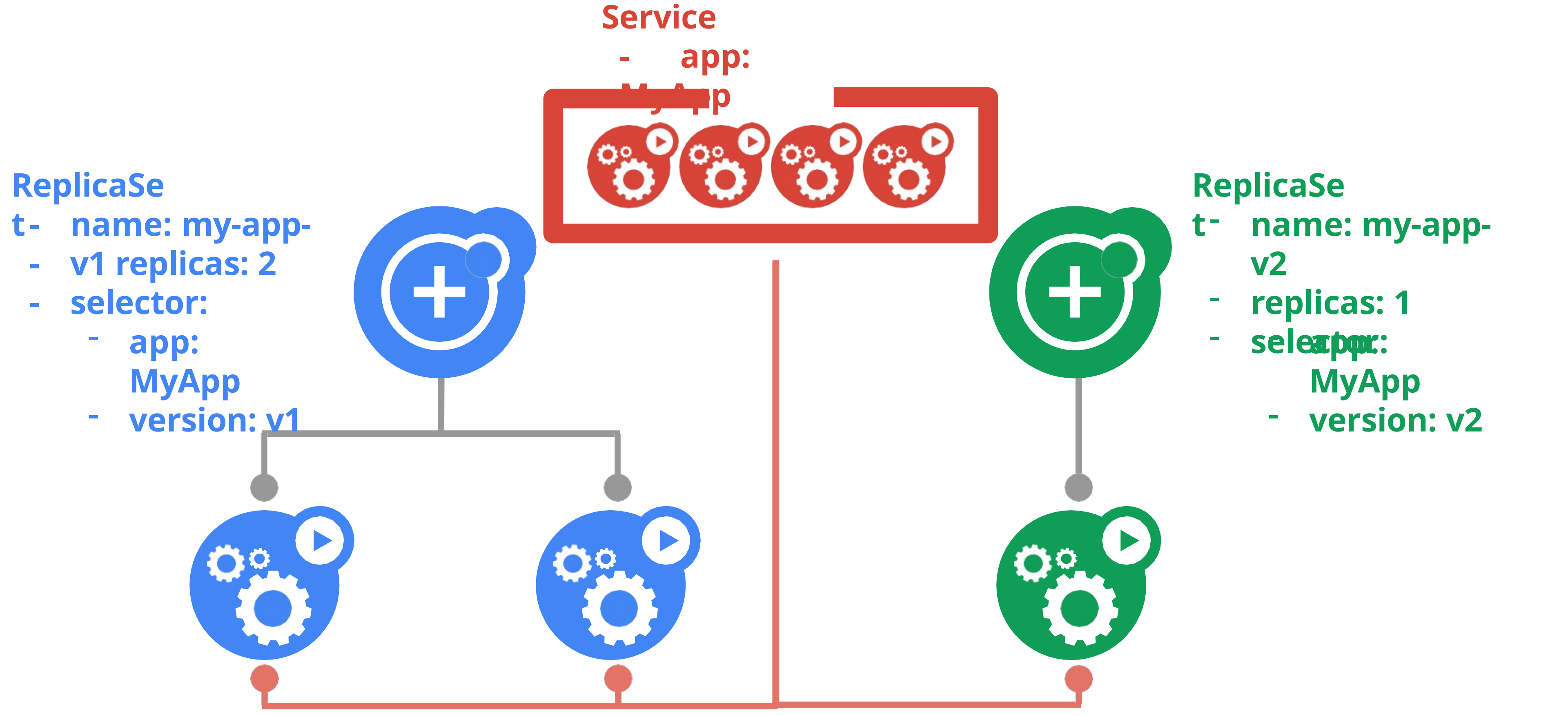
ReplicaSet
- name: my-app-v2
- replicas: 0
- selector:
  - app: MyApp
  - version: v2

Rolling Update



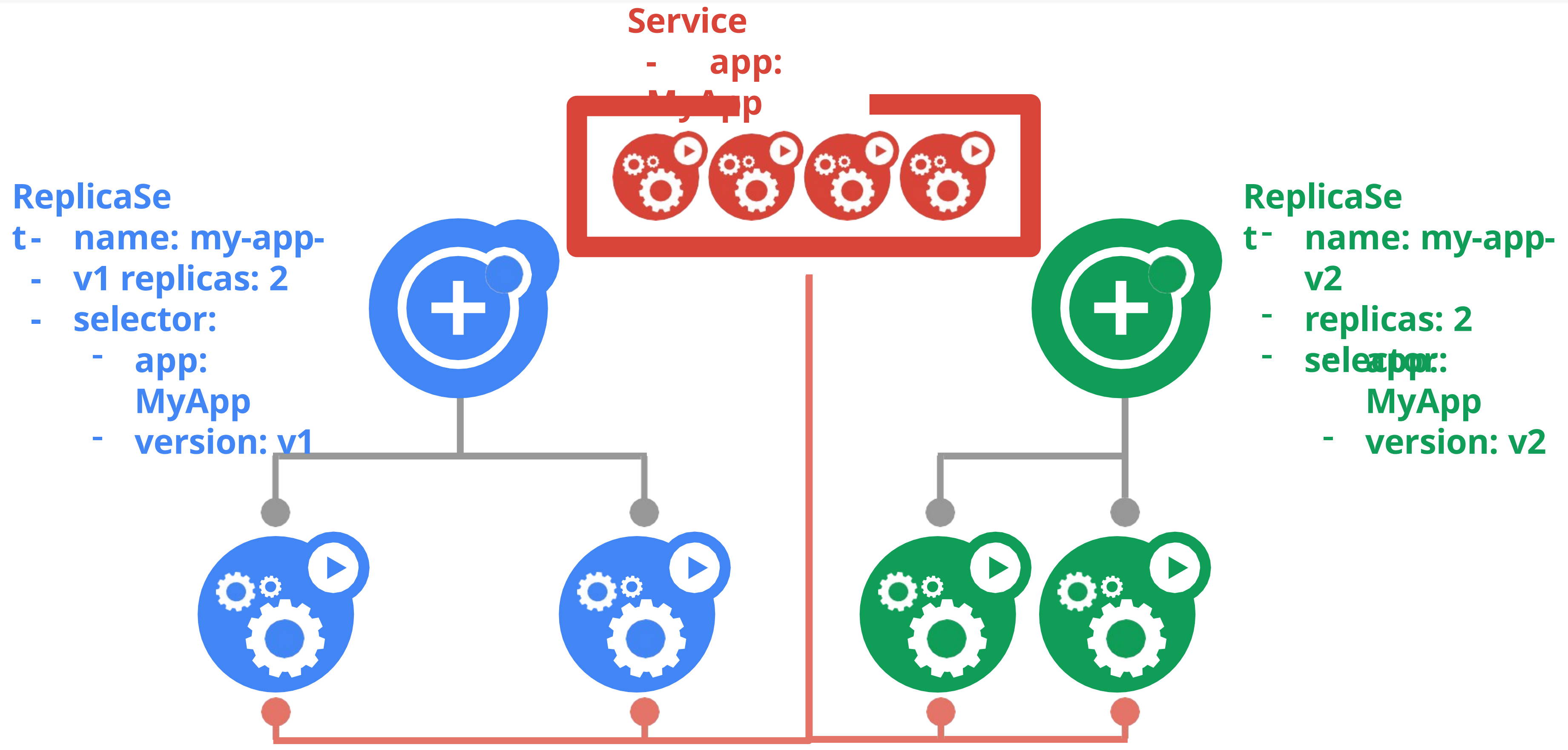
Rolling Update

@meteatamel



Rolling Update

@meteatamel

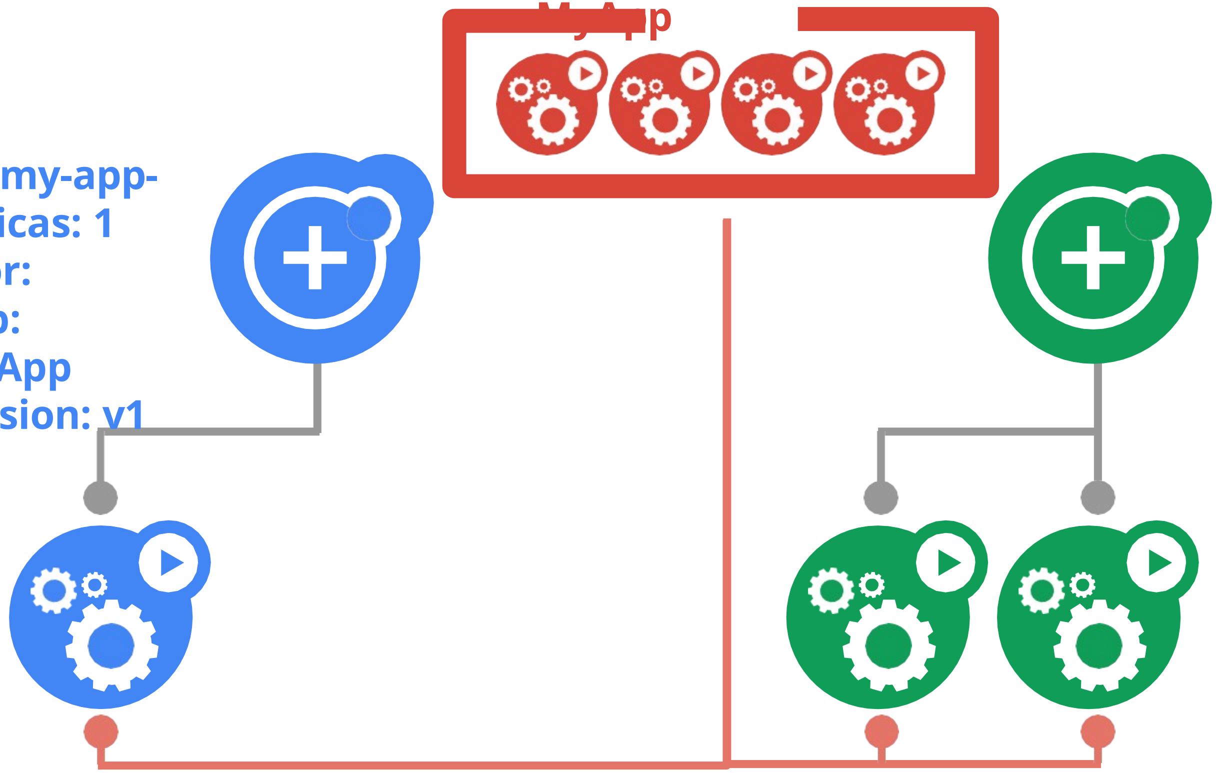


Rolling Update

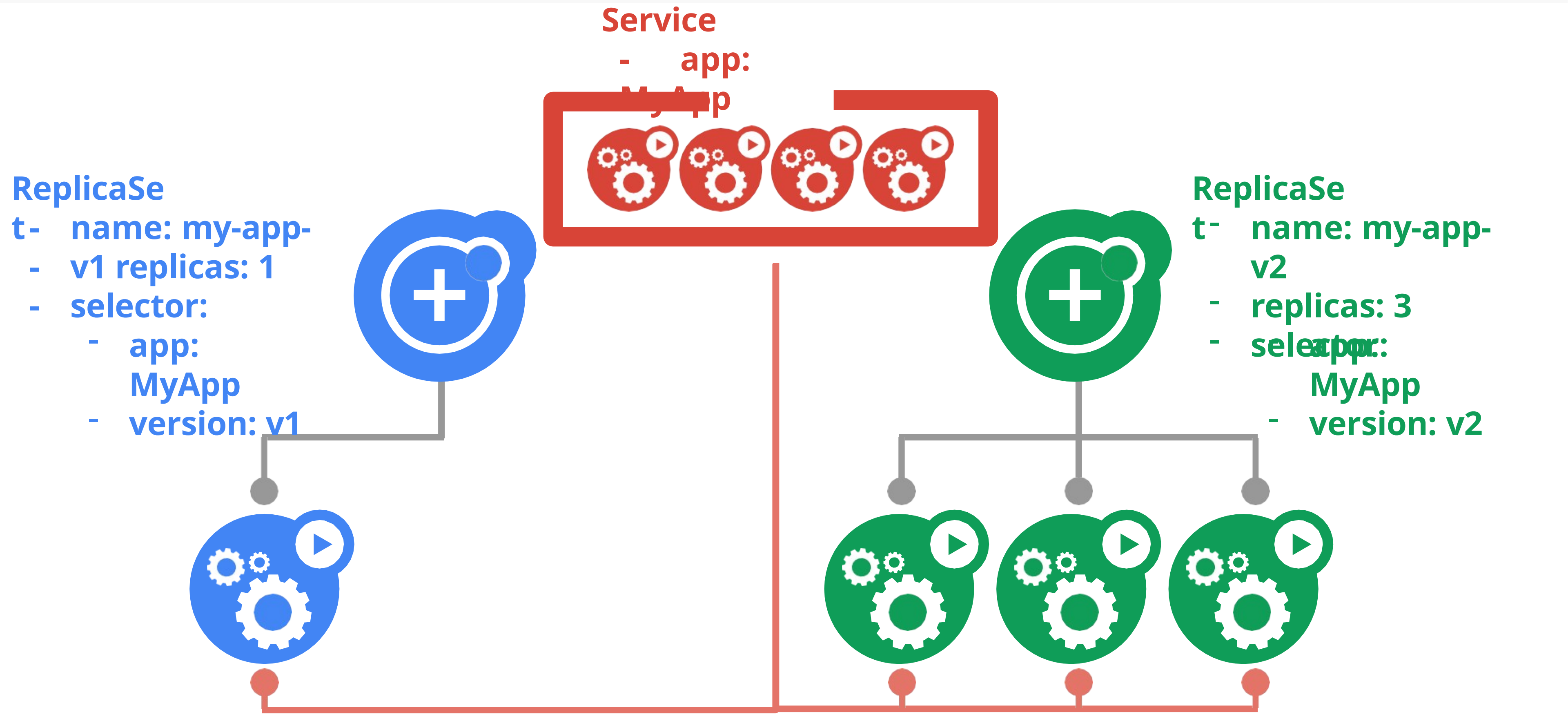
Service
- app: MyApp

ReplicaSet
- name: my-app-v1
- replicas: 1
- selector:
  - app: MyApp
  - version: v1

ReplicaSet
- name: my-app-v2
- replicas: 2
- selector:
  - app: MyApp
  - version: v2

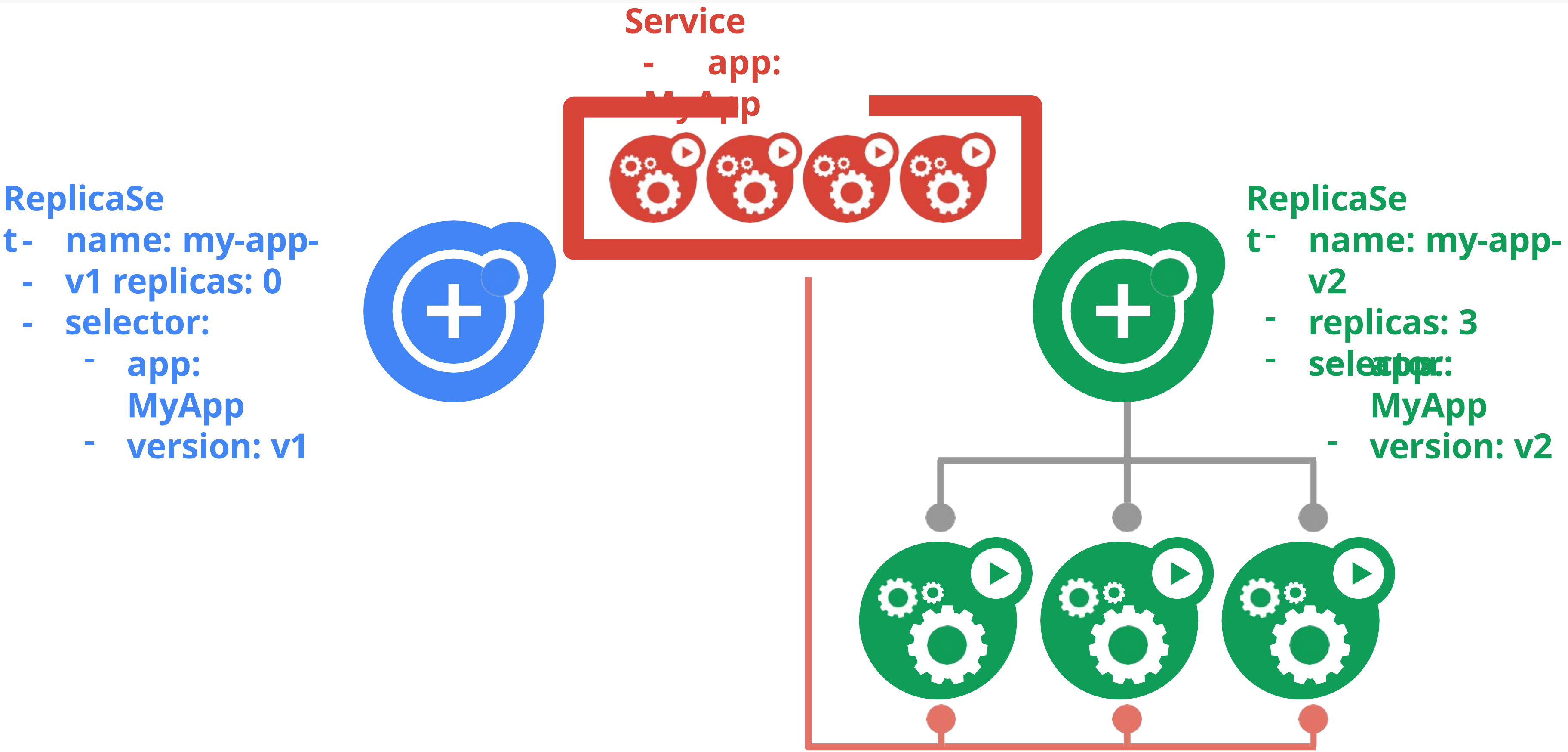


Rolling Update

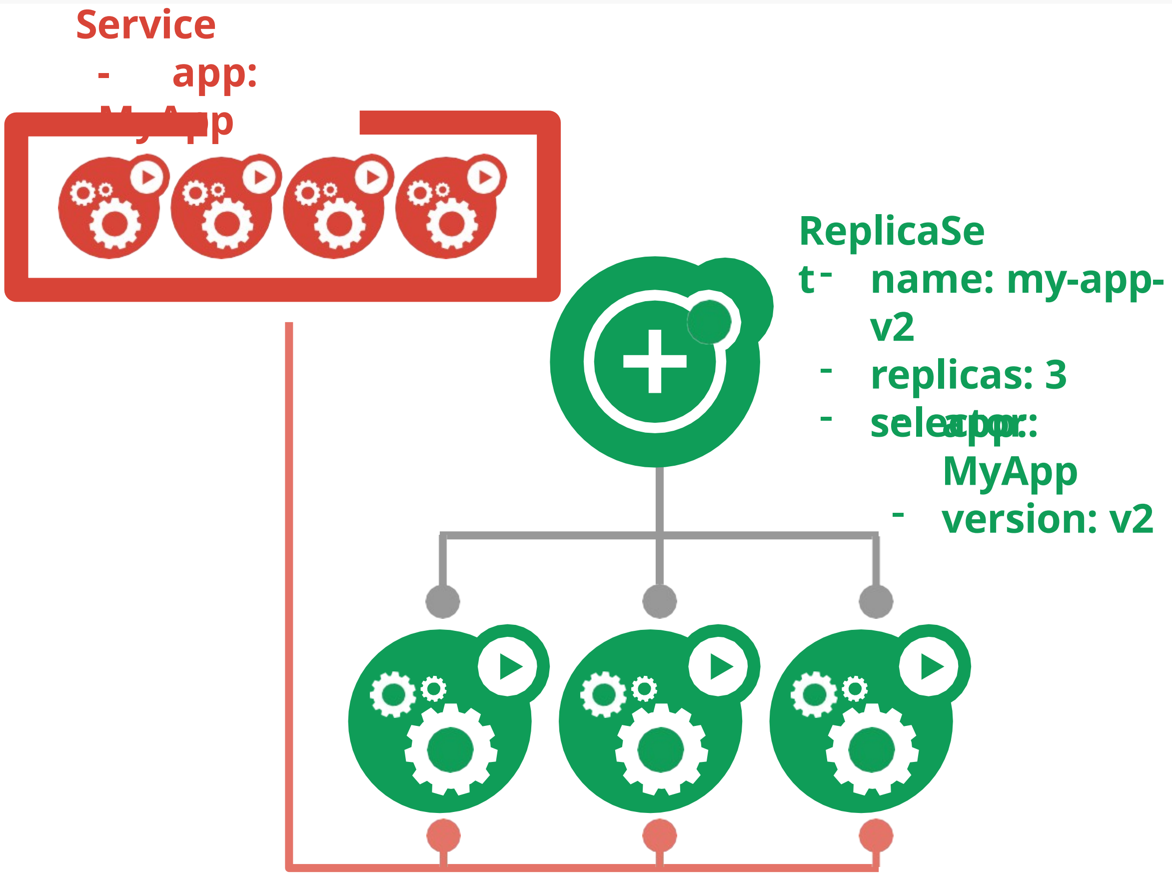


Rolling Update

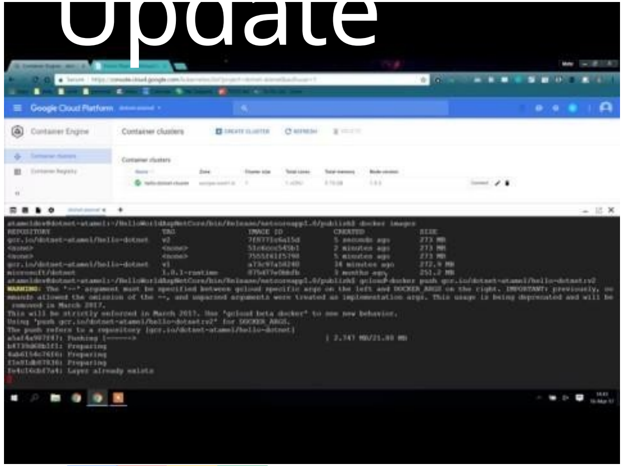
@meteatamel



Rolling Update

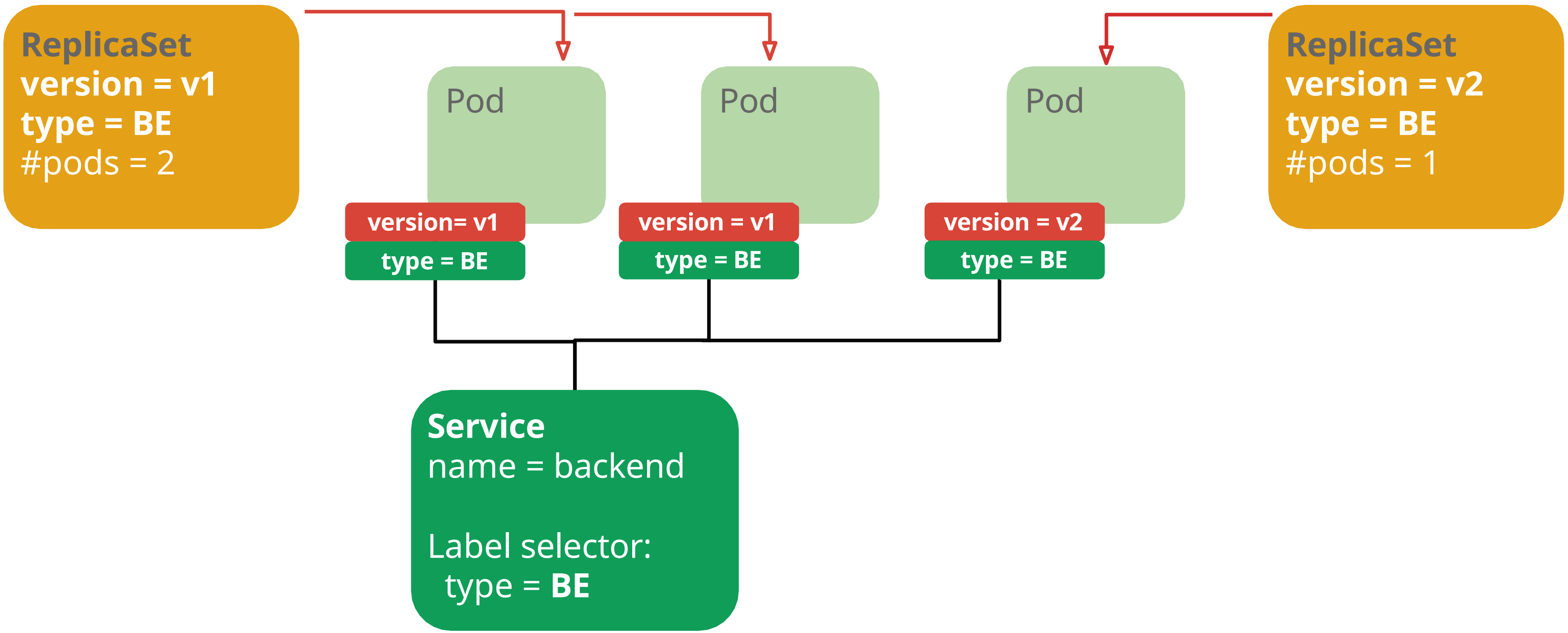


Demo: Rolling Update



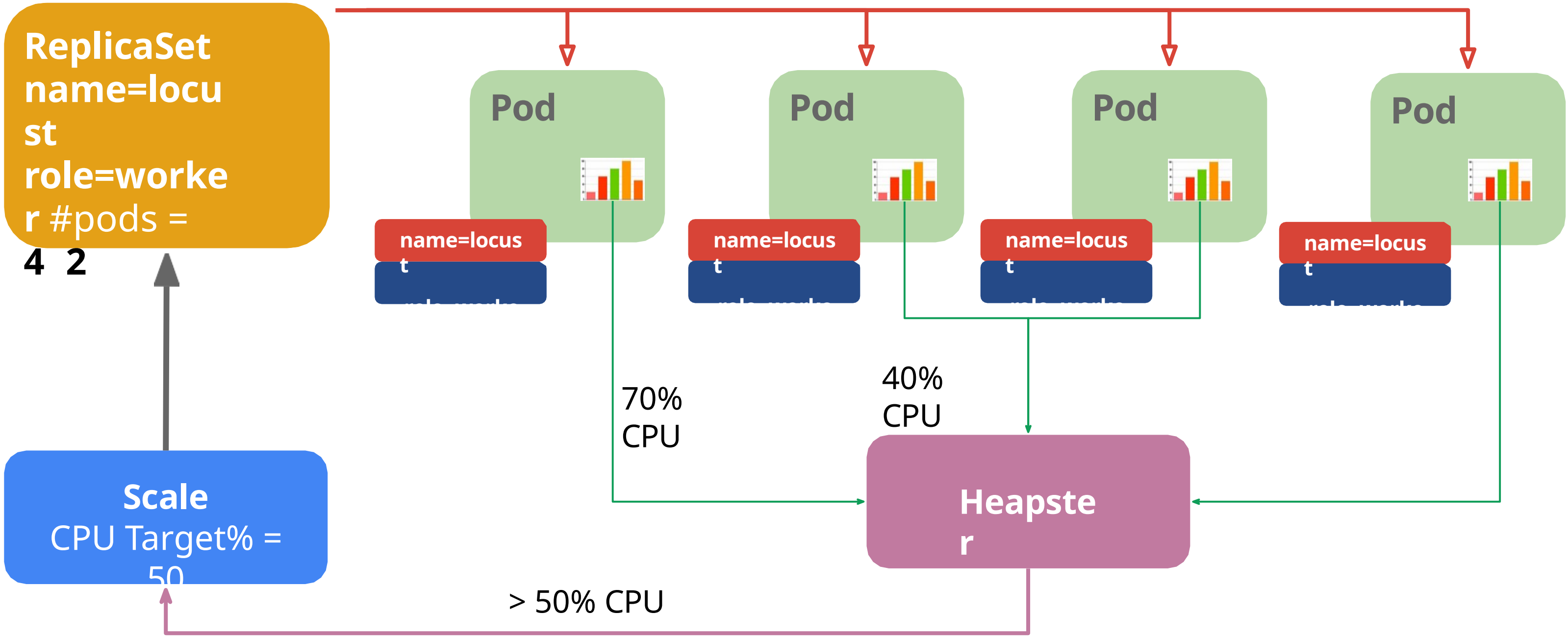
Canary Deployments

@meteatamel



Autoscaling

@meteatame
l



DaemonSet

DaemonSets

Problem: how to run a Pod on every node?

- or a subset of nodes

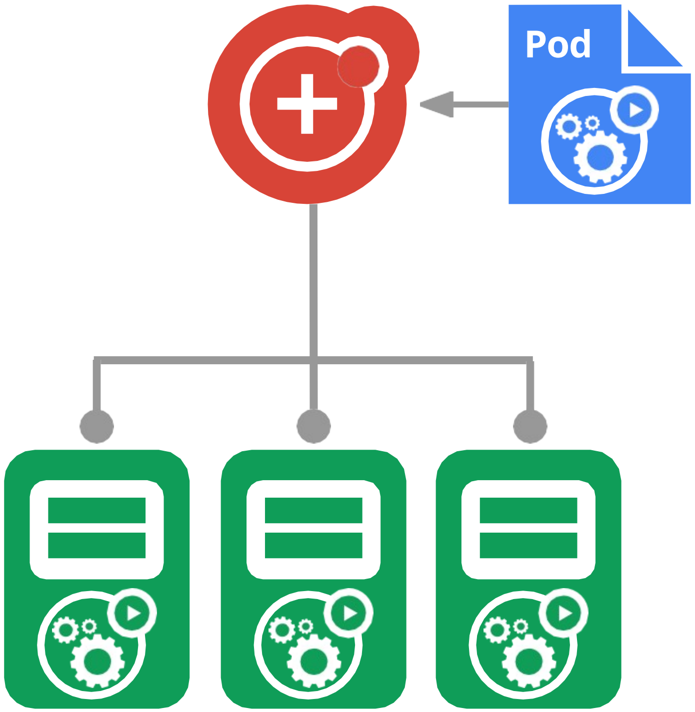
Similar to ReplicaSet

- principle: do one thing, don't overload

"Which nodes?" is a selector

Use familiar tools and

patterns



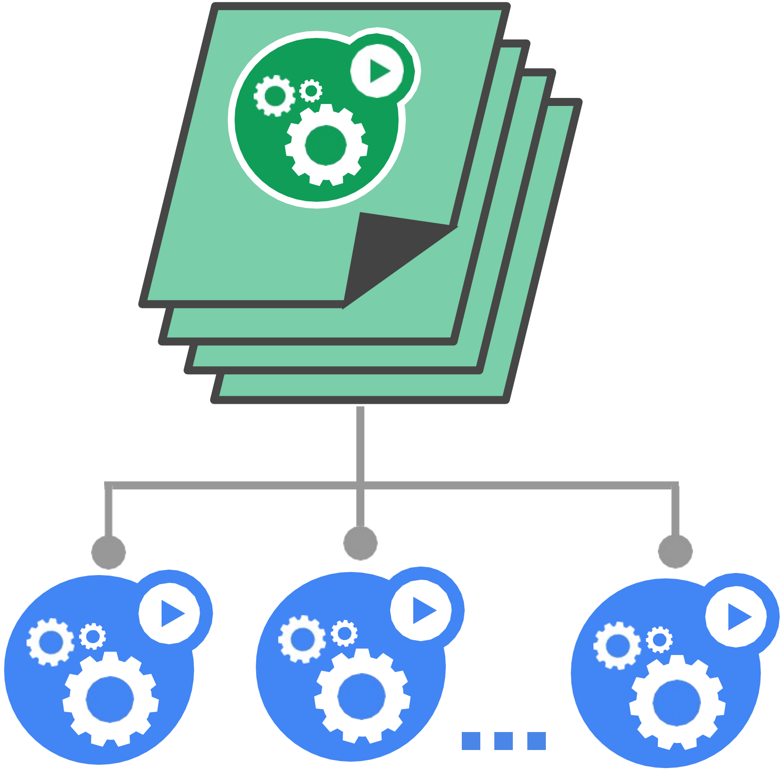
Jobs

Jobs

Run-to-completion, as opposed to run-forever

- Express parallelism vs. required completions
- Workflow: restart on failure
- Build/test: don’t restart on failure

Aggregates success/failure counts Built for batch and big-data work



StatefulSet

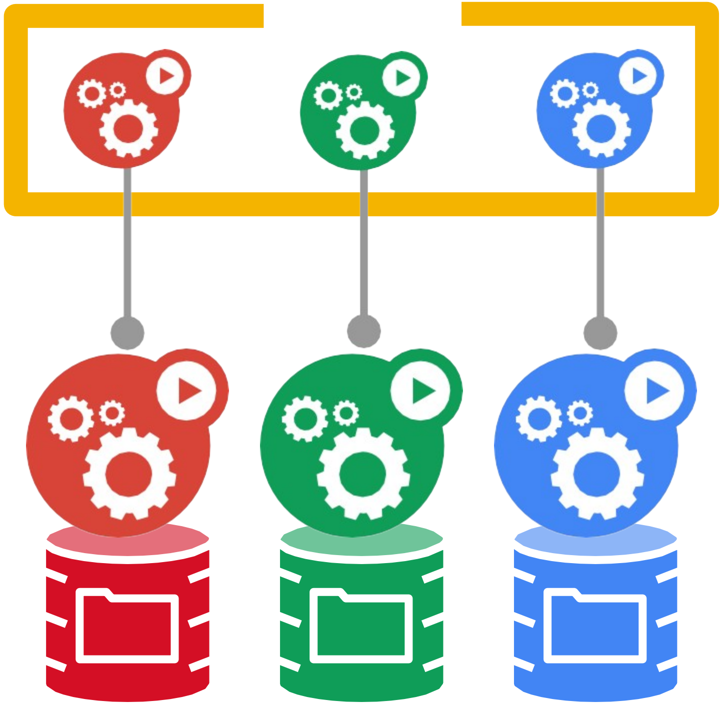
StatefulSets

Goal: enable clustered software on Kubernetes

- mysql, redis, zookeeper, …

Clustered apps need "identity" and sequencing guarantees

- stable hostname, available in DNS
- an ordinal index
- stable storage: linked to the ordinal & hostname
- discovery of peers for quorum
- startup/teardown ordering



ConfigMap

ConfigMaps

@meteatamel

Goal: manage app configuration

- ...without making overly-brittle container images

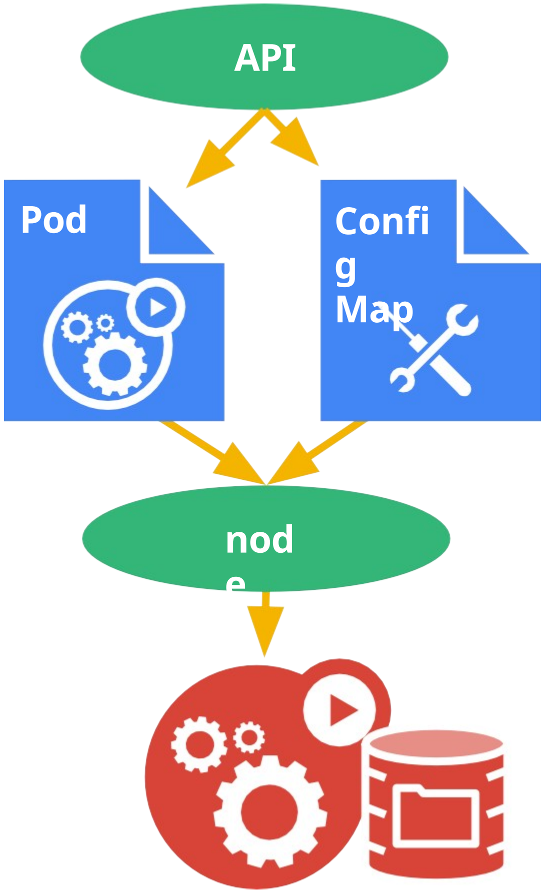
12-factor says config comes from the environment

- Kubernetes is the environment

Manage config via the Kubernetes API

Inject config as a virtual volume into your Pods

- late-binding, live-updated (atomic)
- also available as env vars



Secrets

Secrets

Goal: grant a pod access to a secured *something*

- don’t put secrets in the container image!

12-factor says config comes from the environment

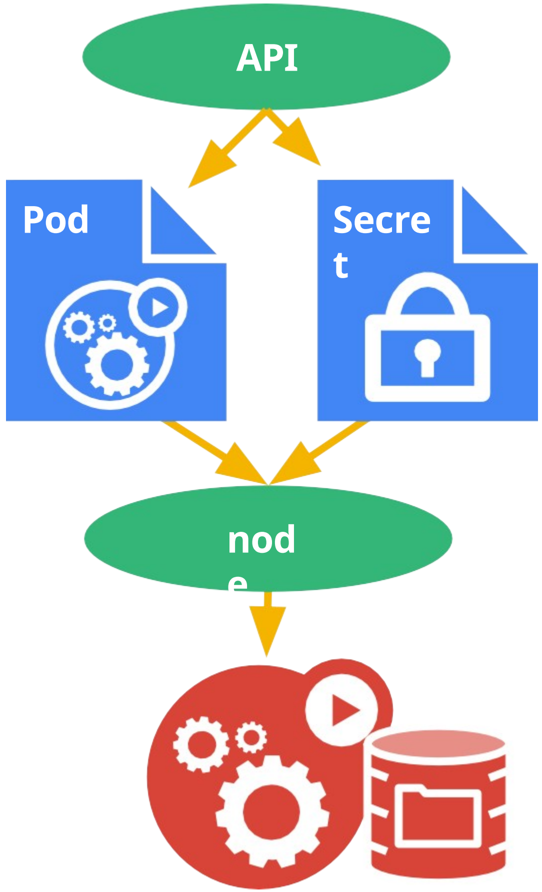
- Kubernetes is the environment

Manage secrets via the Kubernetes API

- late-binding, tmpfs - never touches disk

Inject secrets as virtual volumes into your Pods

- also available as env vars



Kubernetes Terminology

Deployment	**ReplicaSet**	**DaemonSet**
Pod	**Liveness Probe**	**Job**
Volume	**Readiness Probe**	**StatefulSet**
Label	**Service**	**ConfigMap**
Selector		**Secret**

There is more!

@meteatamel

Thank You

kubernetes.io

cloud.google.com/container-engine

Send talk feedback

bit.ly/atamel

Mete Atamel
@meteatamel
atamel@google.com
meteatamel.wordpress.com